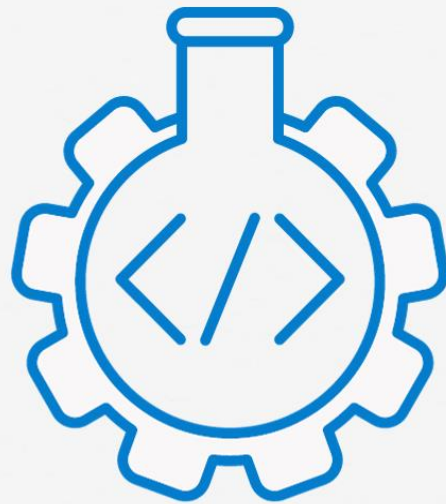
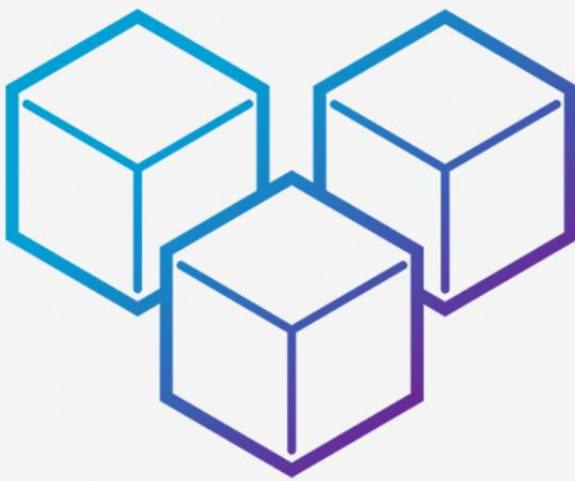


Microservices Certification Training Lab Guide



Microservices Training: Coding Lab Guide

Our learn-by-doing method enables you to build practical/coding experience that sticks. 95% of our learners say they remember more when they learn by doing hands-on which is required to translate knowledge into real-world results. The recommended hardware and software requirements for our training are given below:

Hardware Requirements

- **Processor:** 64-bit processor with 2.5 GHz or faster; Intel i5 or AMD Ryzen 5 or higher.
- **Memory:** 8 GB or 16 GB RAM recommended for smooth multitasking.
- **Storage:** At least 256 GB SSD recommended for faster boot times and data access.
- **Display:** Full HD display with a resolution of 1920 x 1080 or higher recommended.

Software Requirements

- **Supported OS:** Windows 10, Windows 11
- **Not Supported OS:** Windows7, Windows8, Windows 8.1
- Visual Studio 2022 Community or higher
- Latest Visual Studio Code (Optional)
- SQL Server Management Studio
- SQL Server 2022 Express or higher
- WSL2 For Windows
- Docker Desktop
- Git For Windows
- Azure Account For AKS

Software Download Links

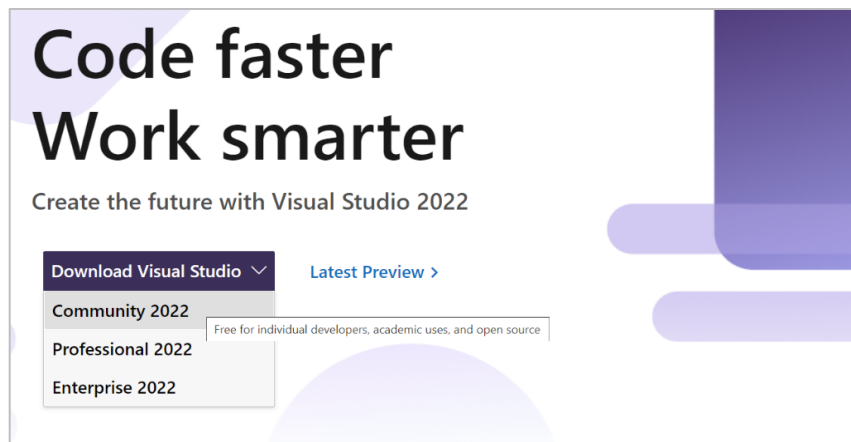
- Visual Studio 2022 Community: <https://visualstudio.microsoft.com/>
- SQL Server 2022 Express: <https://go.microsoft.com/fwlink/p/?linkid=2216019>
- SQL Server Management Studio: <https://aka.ms/ssmsfullsetup>
- WSL2 For Windows 64: <https://wslstorestorage.blob.core.windows.net/wslblob>
- Docker Desktop: <https://www.docker.com/products/docker-desktop>
- Visual Studio Code: <http://code.visualstudio.com/>
- Git (for Windows): <https://git-scm.com/download/win>

1. Visual Studio 2022 Installation

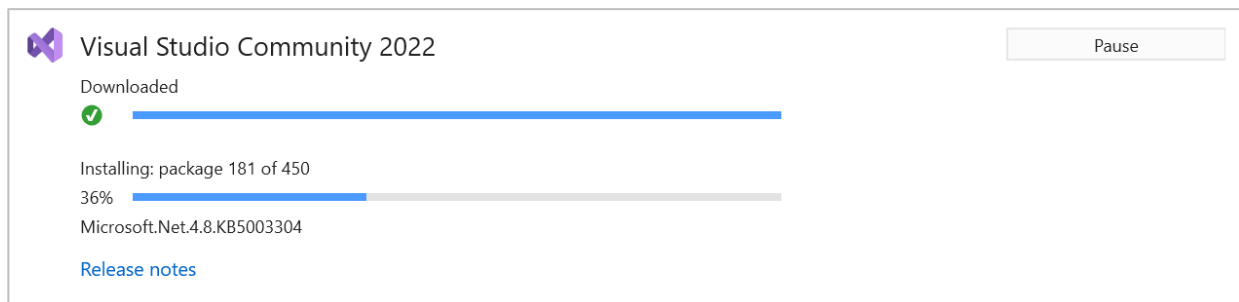
Visual Studio is an Integrated Development Environment (IDE) developed by Microsoft to develop GUI (Graphical User Interface), Web applications, web apps, mobile apps, cloud, and web services, etc.

To install and use Visual Studio for commercial purposes one must buy a license from Microsoft. For learning (non-commercial) purposes, Microsoft provided a free Visual Studio Community Version. We will use the Visual Studio Community Version 2022.

- **Step 1: Download the** <https://visualstudio.microsoft.com/downloads/>

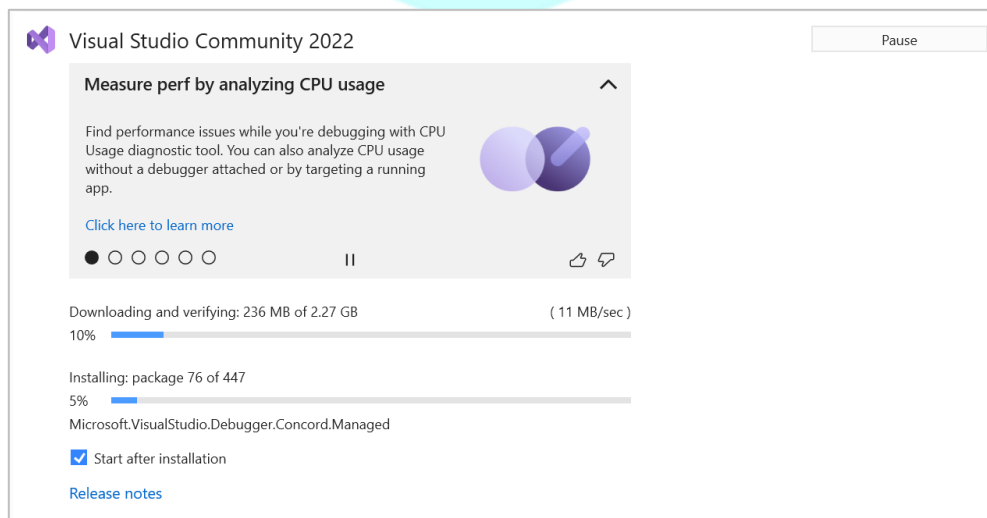
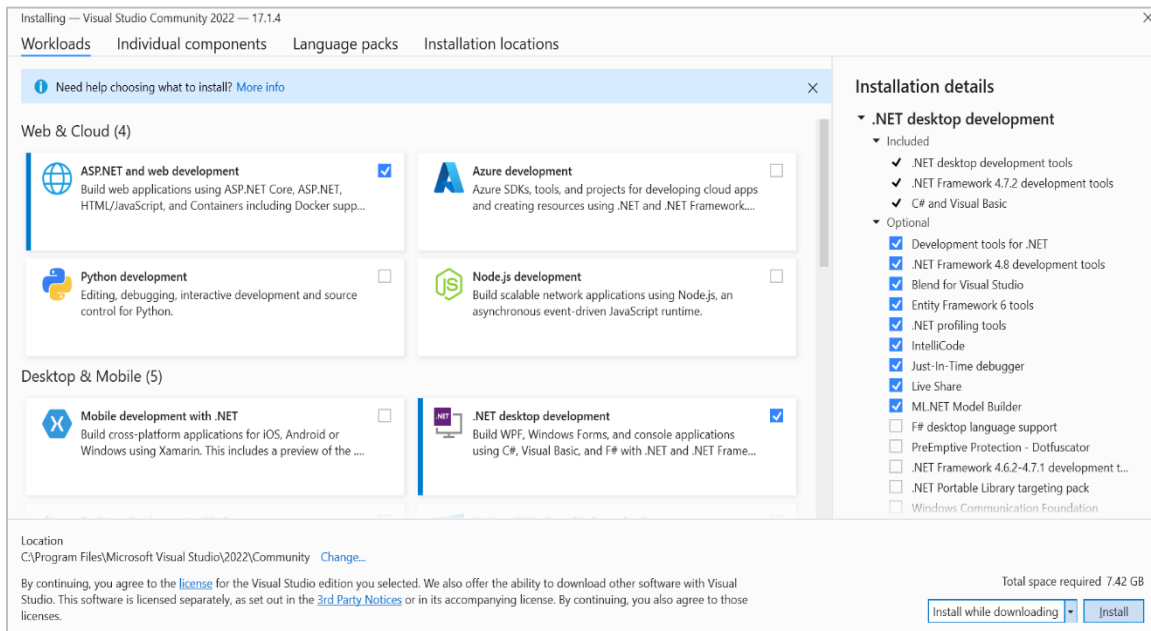


- **Step 2: Run the .exe file and follow the instructions to install the Visual Studio Community Version on the system.**



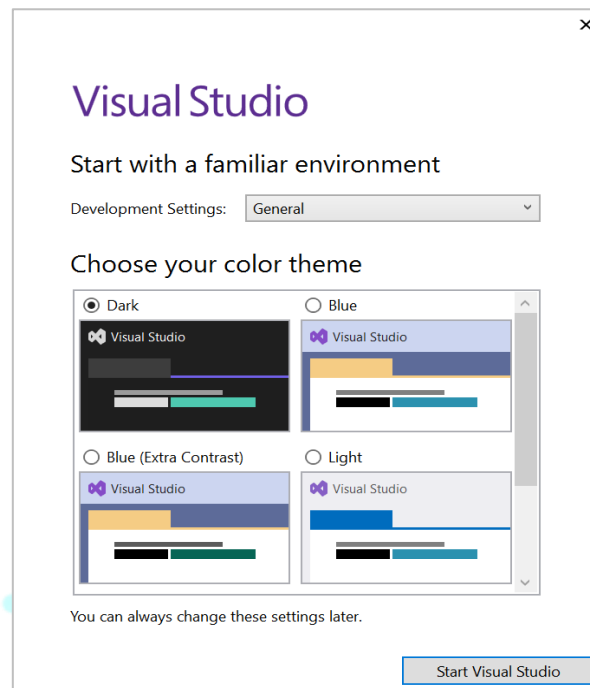
- **Step 3: Select ASP.NET and web development and .NET desktop development for windows forms and console applications etc. from the options and click to install in the bottom right corner as shown below.**

We are selecting both options. We can also modify it after installation.

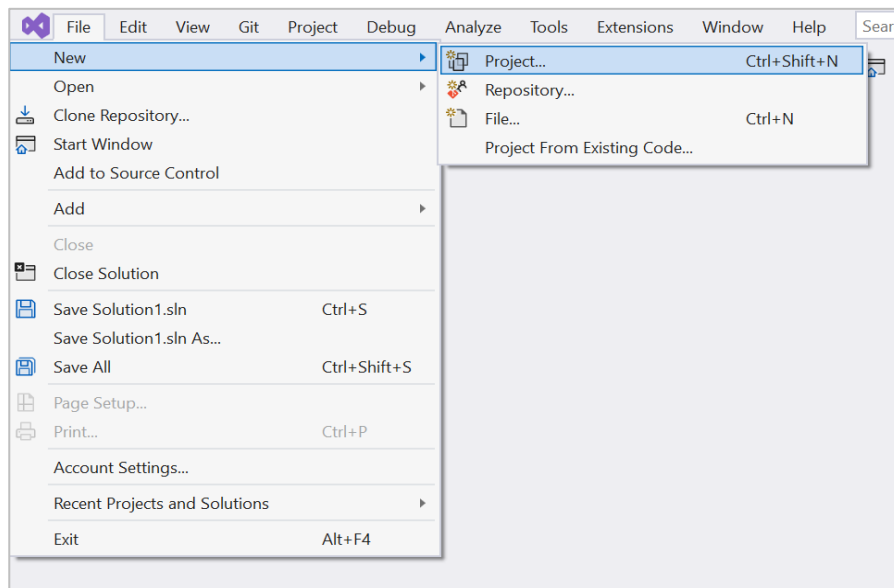


- **Step 4: Click on launch and then it displays the sign-in step which is completely optional.** The dialog box will appear to choose the Development Settings and color theme. You need not to change the development settings.

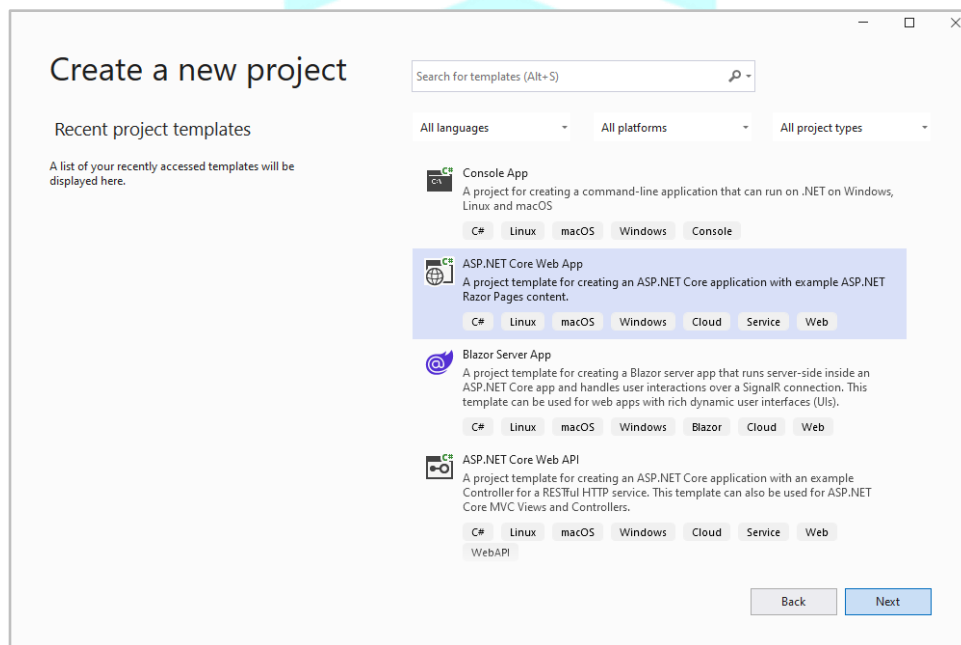
Once select the required options, click on the Start Visual Studio option.



- **Step 5: To create a new ASP.NET Web application, Go to File → New →Project like as shown below:**

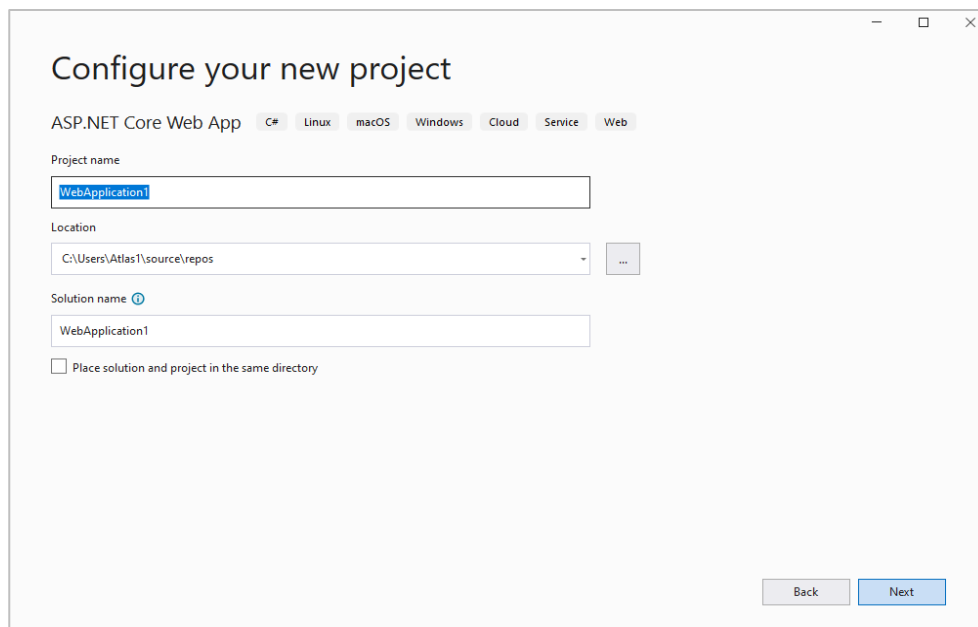


- **Step 6: As soon as we select the project, we will notice the different options of the Project. Select ASP.NET Core Web App and click *Next*.**



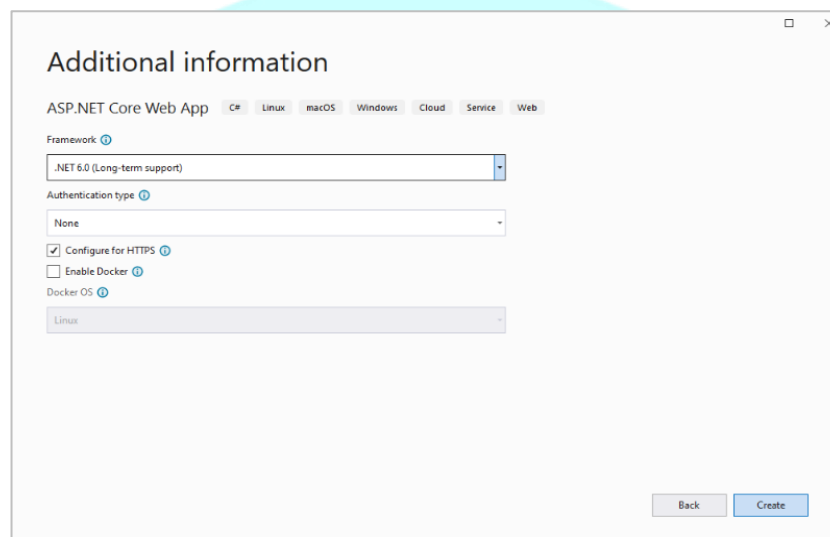
- **Step 7: The next step is to configure the project. Here, we have to choose the Project Name and Solution name and click on Next Button.**

We can also change the location of the project. We can put a different name for the solution as the solution is like a container for projects. We are putting the Project name and Solution name as WebApplication1 as shown in the below screenshot.



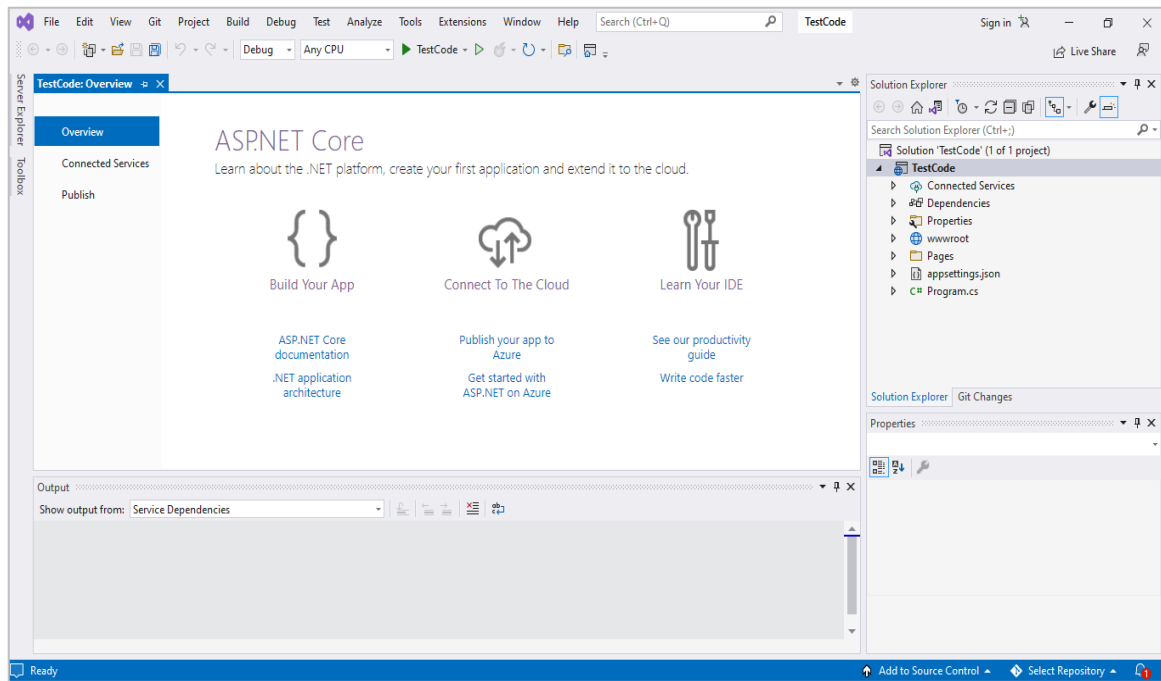
After Clicking the Next button, the Additional Information dialog box appears.

You don't have to modify anything. Just click on Create button.



- **Step 8: After Clicking on Create, Your Project is being created as shown in the following image.**

Visual Studio has been successfully installed. You're all set!



2. SQL Server 2022 Express Installation

MS SQL Server 2022 is the latest version of Microsoft's SQL Server relational database management system, designed for high-performance, scalable, and secure data management. It offers advanced features for database development, data integration, and analytics. We are going to install SQL Server 2022 on Windows Server 2022.

- **Step 1:** Before we begin, head over to the <https://www.microsoft.com/en-us/sql-server/sql-server-downloads> and download SQL Server 2022.

Or, download a free specialized edition



Developer

SQL Server 2022 Developer is a full-featured free edition, licensed for use as a development and test database in a non-production environment.

[Download now](#)



Express

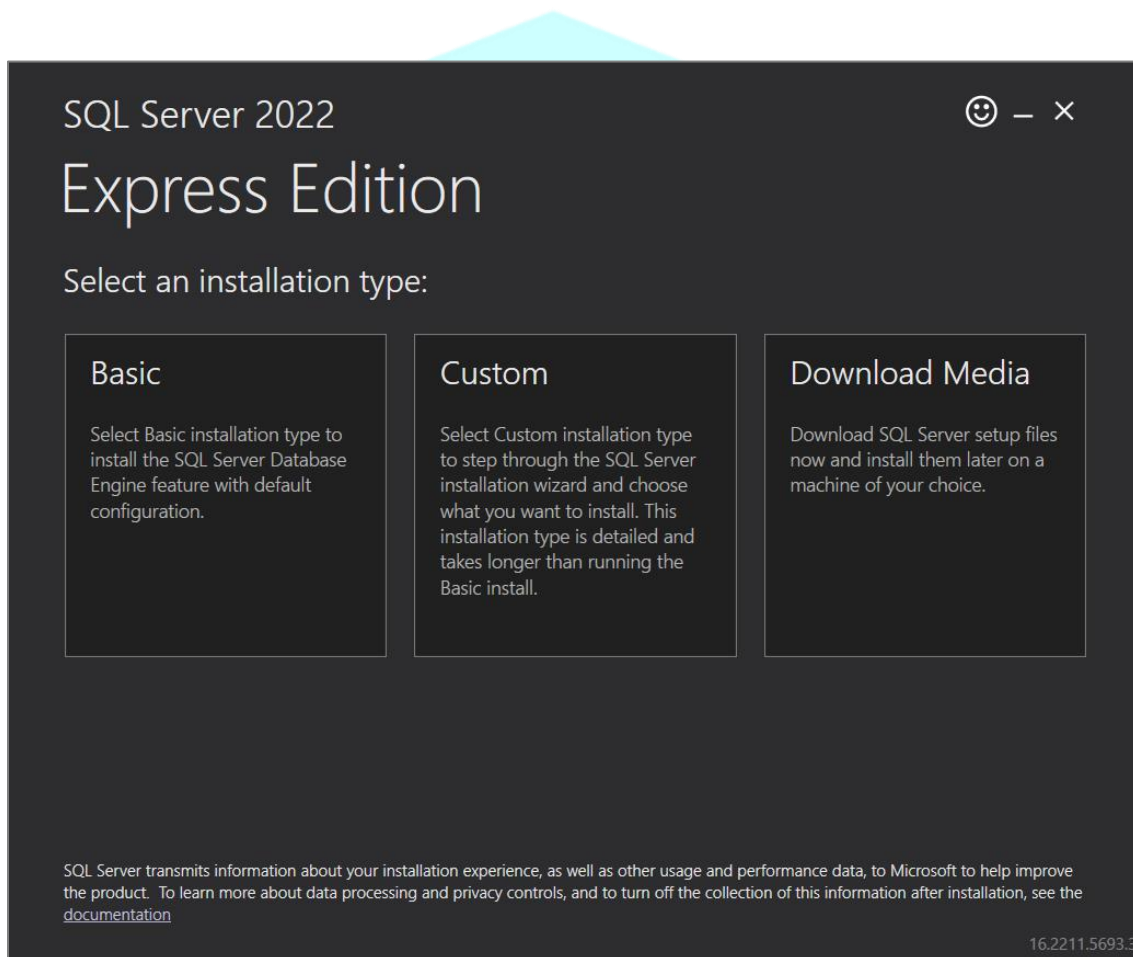
SQL Server 2022 Express is a free edition of SQL Server, ideal for development and production for desktop, web, and small server applications.

[Download now](#)

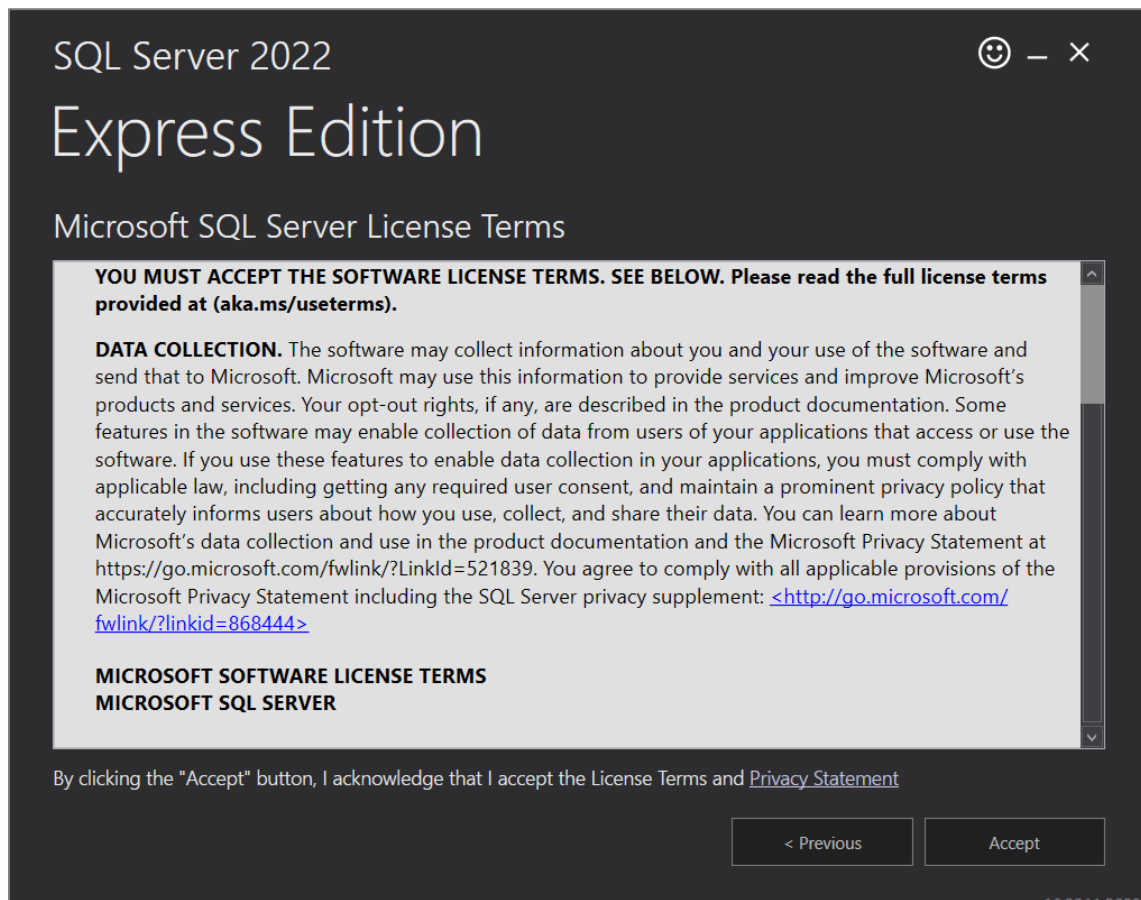
- **Step 2: An .exe file will be downloaded to your system. Double-click that file for further installation.**



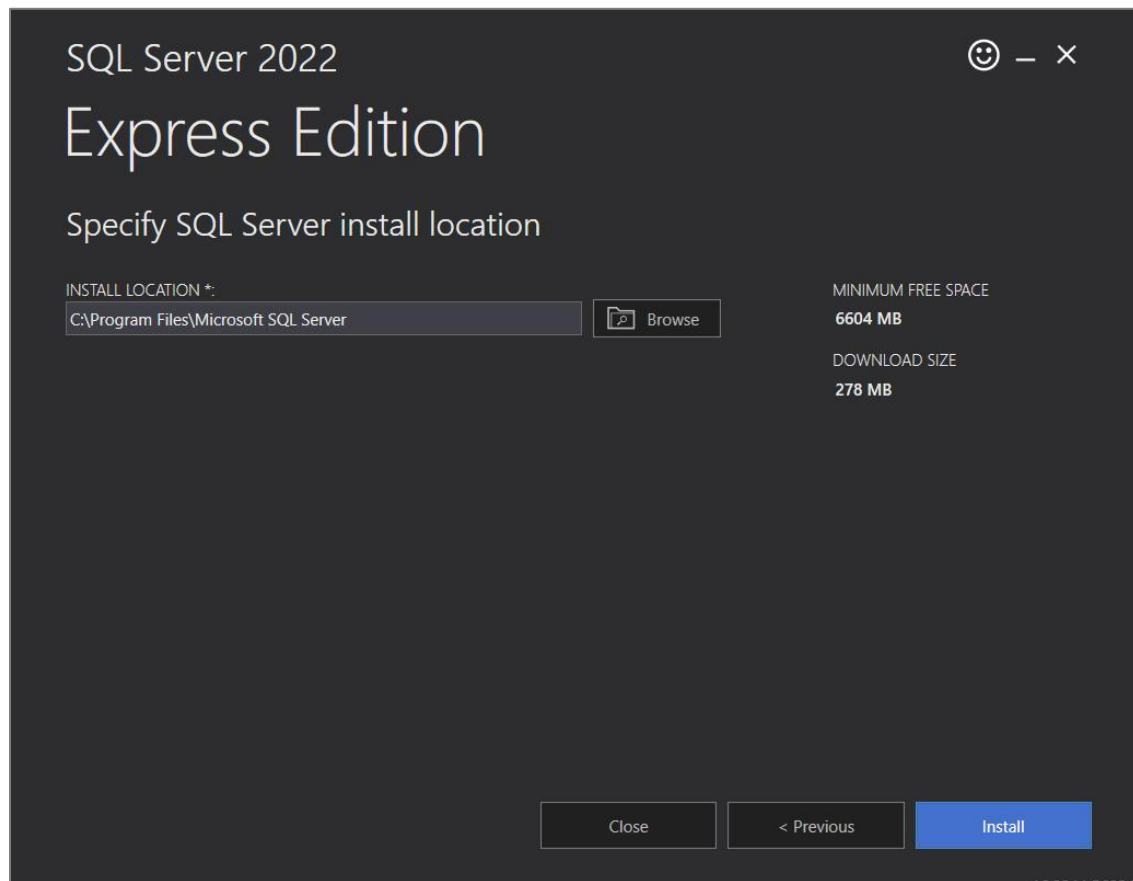
- **Step 3: After double-clicking that .exe file, we have to choose our installation type.**
Let's start with Basic Type as it will install default packages by downloading them from the internet whereas Custom Type provides customized features of SQL Server.



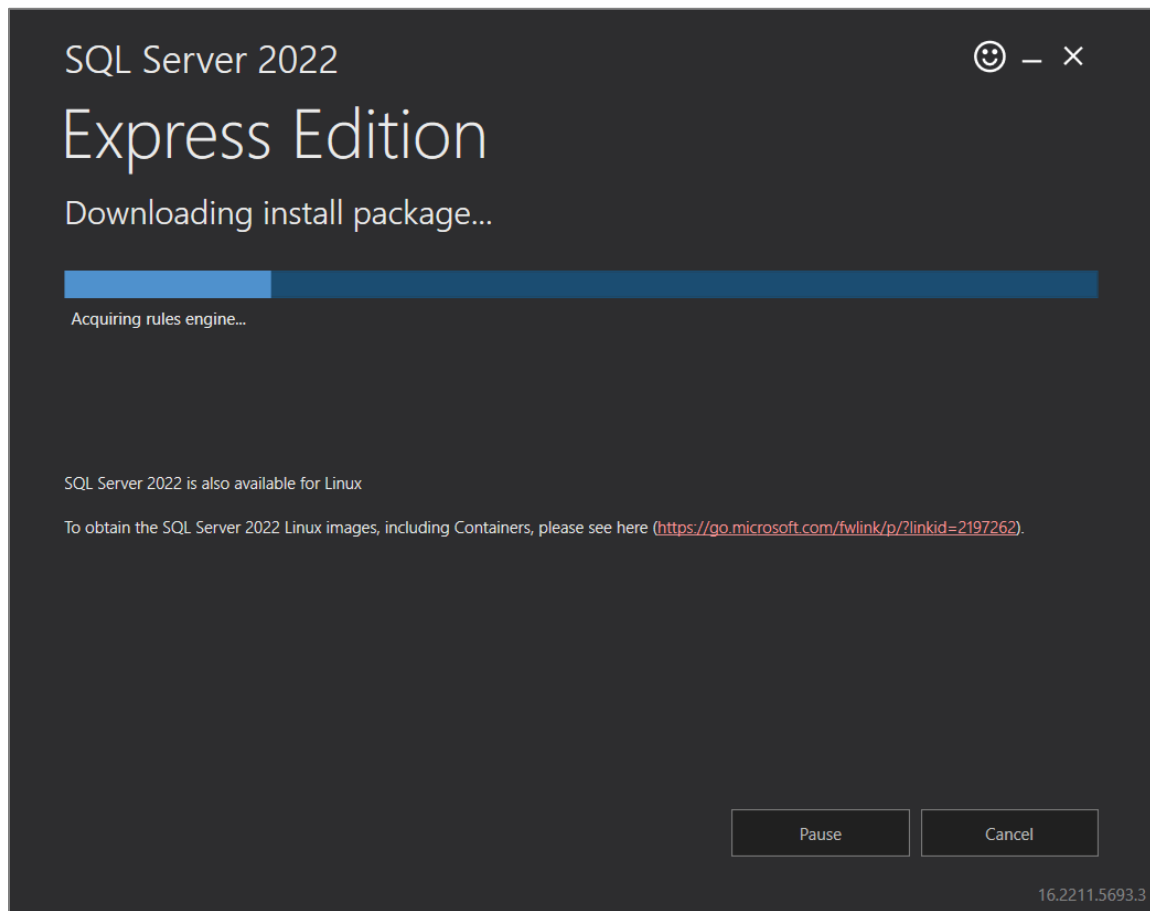
Step 4: Accept the License Agreement.



- Step 5: After that, Specify the location for the install package and click install.

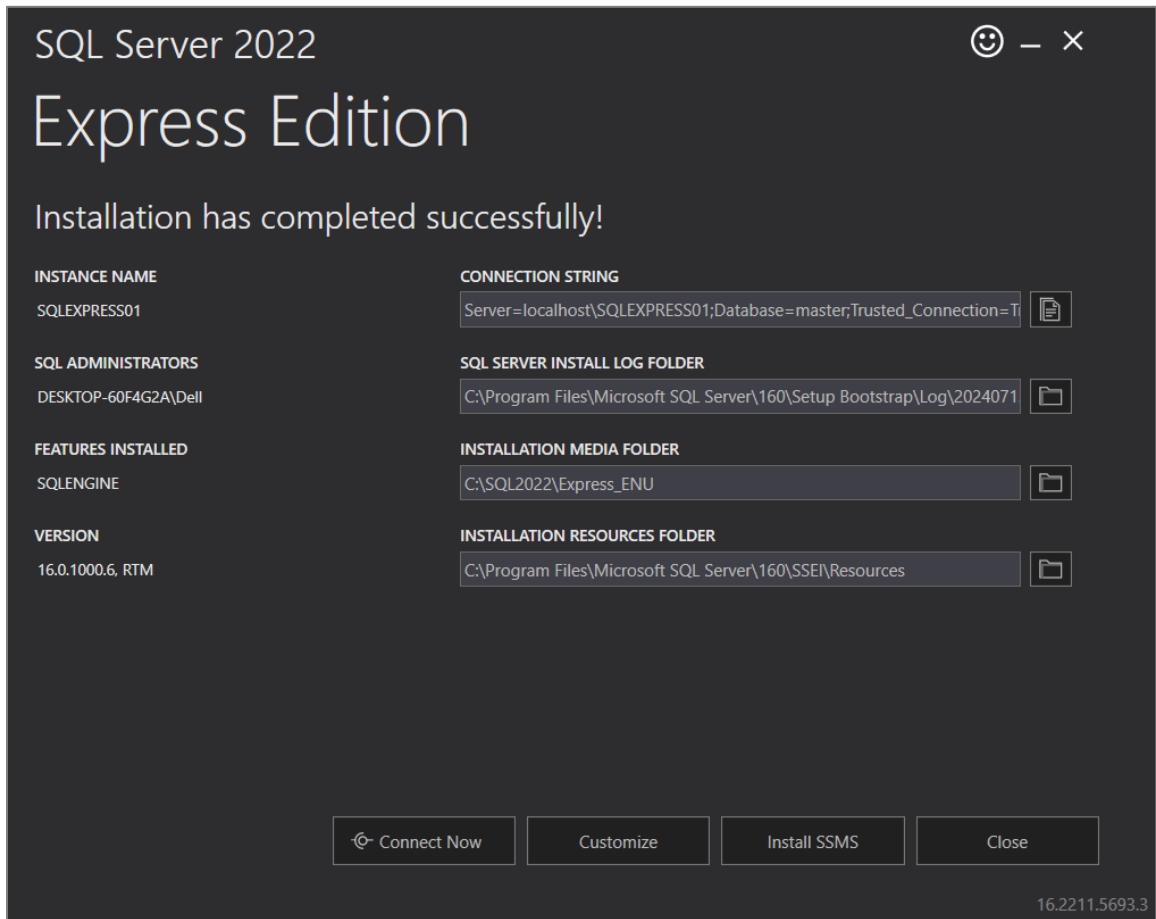


Step 6: After launching the installation process, you will see a progress bar informing you that the install package is being downloaded and that SQL Server Express is being installed.

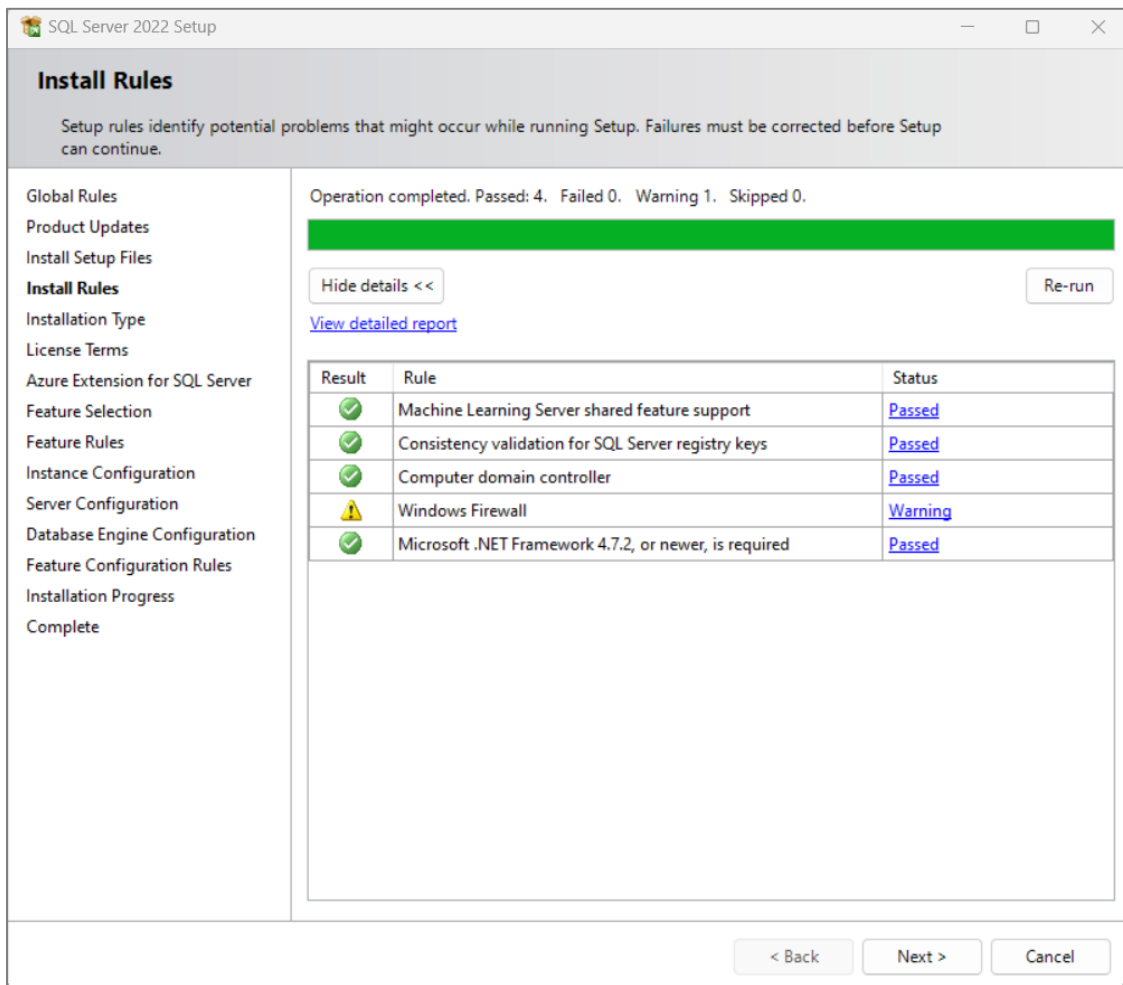


- **Step 7: After the installation has been completed, we can do the following:**
 - Connect to the SQL Server we have just installed via the sqlcmd utility.
 - Launch the SQL Server setup to customize the default configuration for the installation.
 - Install SQL Server Management Studio.
 - Close the installer.

Let's customize or modify the SQL Server 2022 Express Edition configuration we have just installed.



- **Step 8: In the Install Rules step, make sure all rules are passed or contain warnings that do not prevent installation. After checking the checks, click "Next" to proceed.**



- **Step 9: The SQL Server 2022 Setup wizard opens on the Installation Type tab where you choose either to install a new instance or modify the existing one.**
Here, we are adding features to an existing instance of SQL Server 2022. Click 'Next'.

SQL Server 2022 Setup

Installation Type

Perform a new installation or add features to an existing instance of SQL Server 2022.

Global Rules

Product Updates

Install Setup Files

Install Rules

Installation Type

License Terms

Azure Extension for SQL Server

Feature Selection

Feature Rules

Instance Configuration

Server Configuration

Database Engine Configuration

Feature Configuration Rules

Installation Progress

Complete

☒ Perform a new installation of SQL Server 2022

Select this option if you want to install a new instance of SQL Server or want to install shared components.

☐ Add features to an existing instance of SQL Server 2022

SQLEXPRESS01

Select this option if you want to add features to an existing instance of SQL Server. For example, you want to add the Analysis Services features to the instance that contains the Database Engine. Features within an instance must be the same edition.

Installed instances:

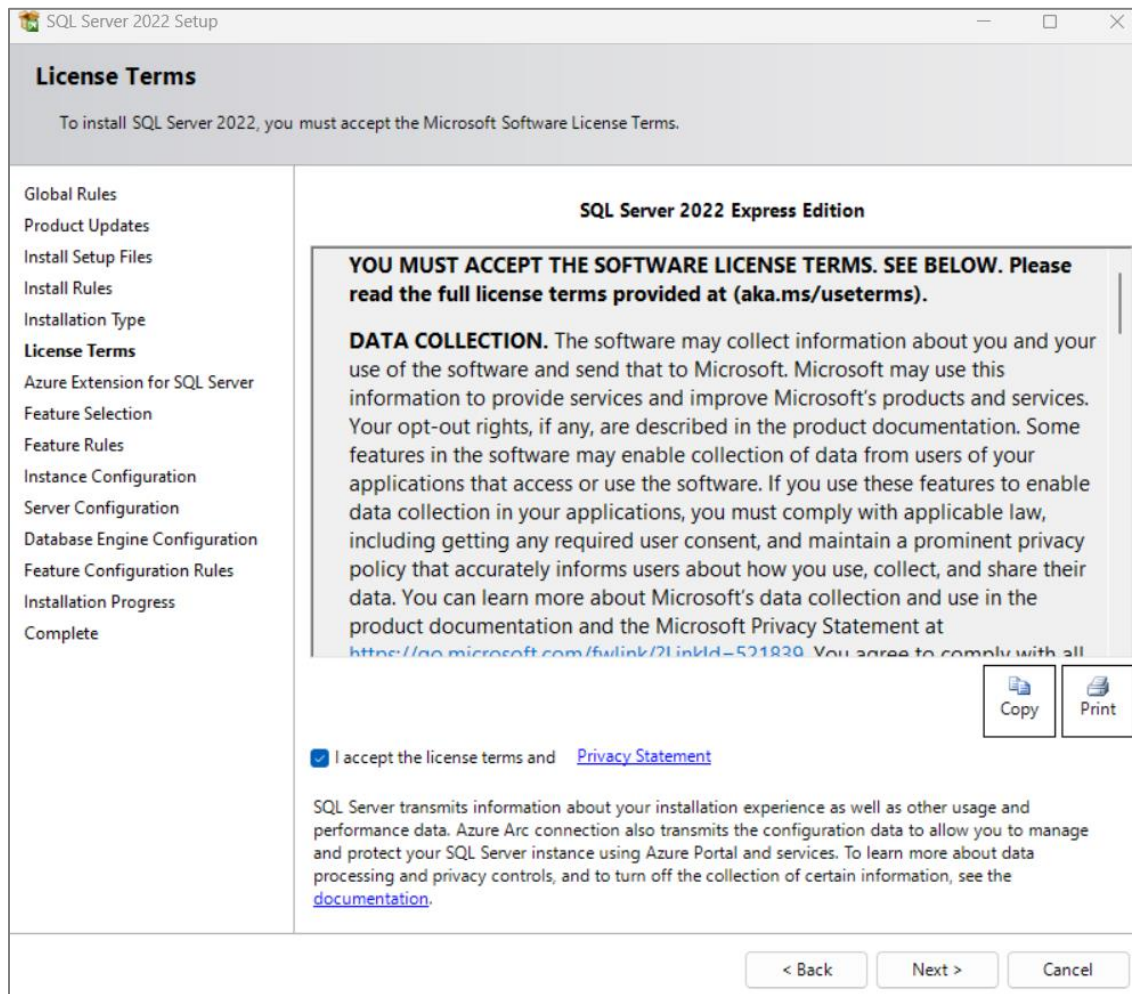
Instance Name	Instance ID	Features	Edition	Version
SQLEXPRESS01	MSSQL16.SQLEXPRESS	SQLEngine	Express	16.0.1000.6
SQLEXPRESS	MSSQL10.SQLEXPRESS	SQLEngine,SQLEn...	Express	10.1.2531.0

< Back

Next >

Cancel

- **Step 10: In the License Terms step, please read the terms and tick the box to accept them. Once approved, click "Next" to continue with the installation.**



- **Step 11: On the Azure Extension for SQL Server, we do not have an Azure account, thus uncheck the "Azure Extension for SQL Server" checkbox. Then click the "Next" button.**

SQL Server 2022 Setup

Azure Extension for SQL Server

Azure Extension for SQL Server is required to enable Microsoft Defender for Cloud, Purview, and Azure Active Directory.

Global Rules
Product Updates
Install Setup Files
Install Rules
Installation Type
License Terms
Azure Extension for SQL Serv...
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Installation Progress
Complete

☐ Azure Extension for SQL Server

To install Azure extension for SQL Server, provide your Azure account or a service principal to authenticate the SQL Server instance to Azure. You also need to provide the Subscription ID, Resource Group, Region, and Tenant ID where this instance will be registered. For more information for each parameter, visit <https://aka.ms/arc-sql-server>.

☐ Use Azure Login

☒ Use Service Principal

Azure Service Principal ID*

Azure Service Principal Secret*

Azure Subscription ID*

Azure Resource Group*

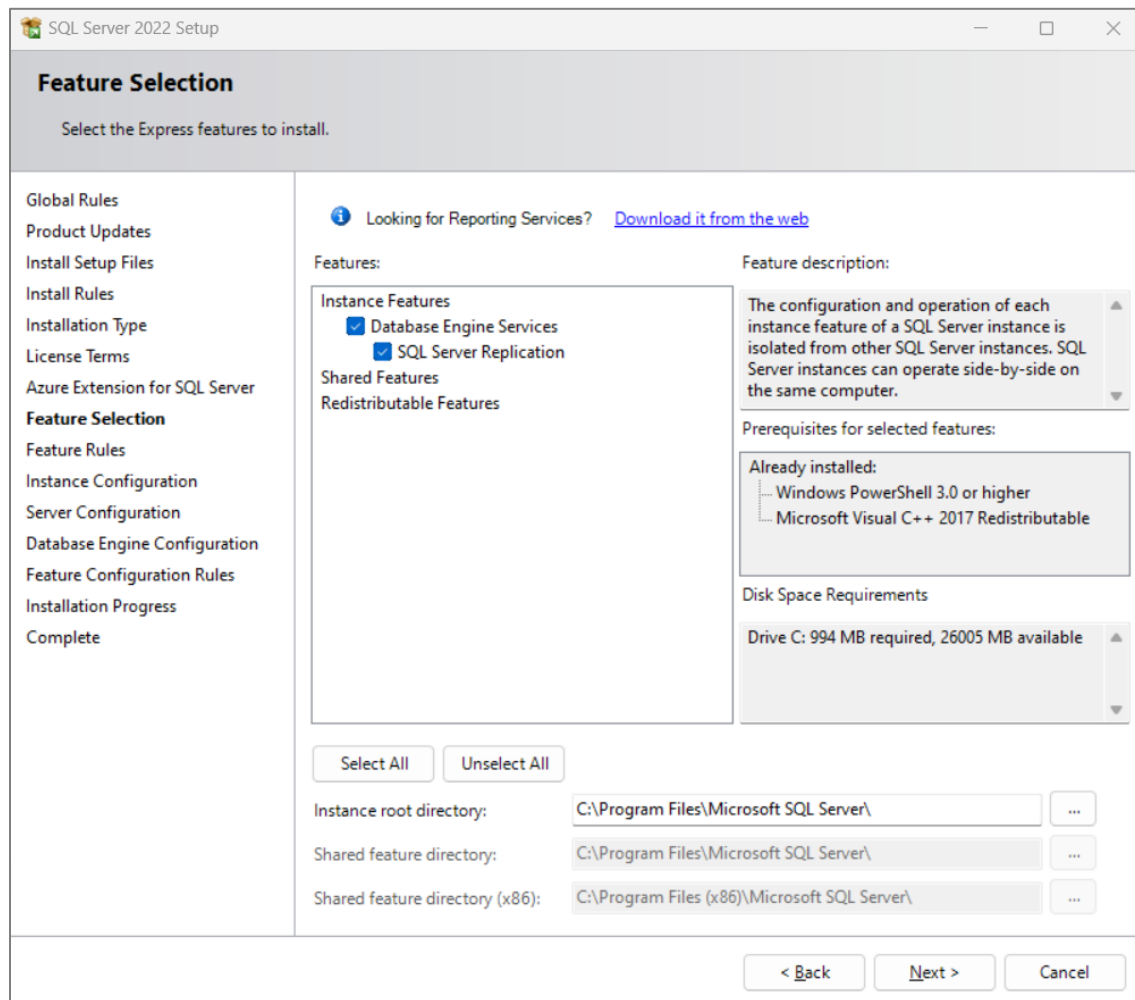
Azure Region*

Azure Tenant ID*

Proxy Server URL (optional)

< Back
Next >
Cancel

- **Step 12: On the Feature Selection tab, select the instance features to be installed. Click Next.**



- **Step 13: In the Instance Configuration step, choose "Named instance" and enter the SQL Server instance name. The name you enter will also be the Instance ID; click "Next" to proceed.**

SQL Server 2022 Setup

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Global Rules
Product Updates
Install Setup Files
Install Rules
Installation Type
License Terms
Azure Extension for SQL Server
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Installation Progress
Complete

☐ Default instance

☒ Named instance: *

Instance ID:

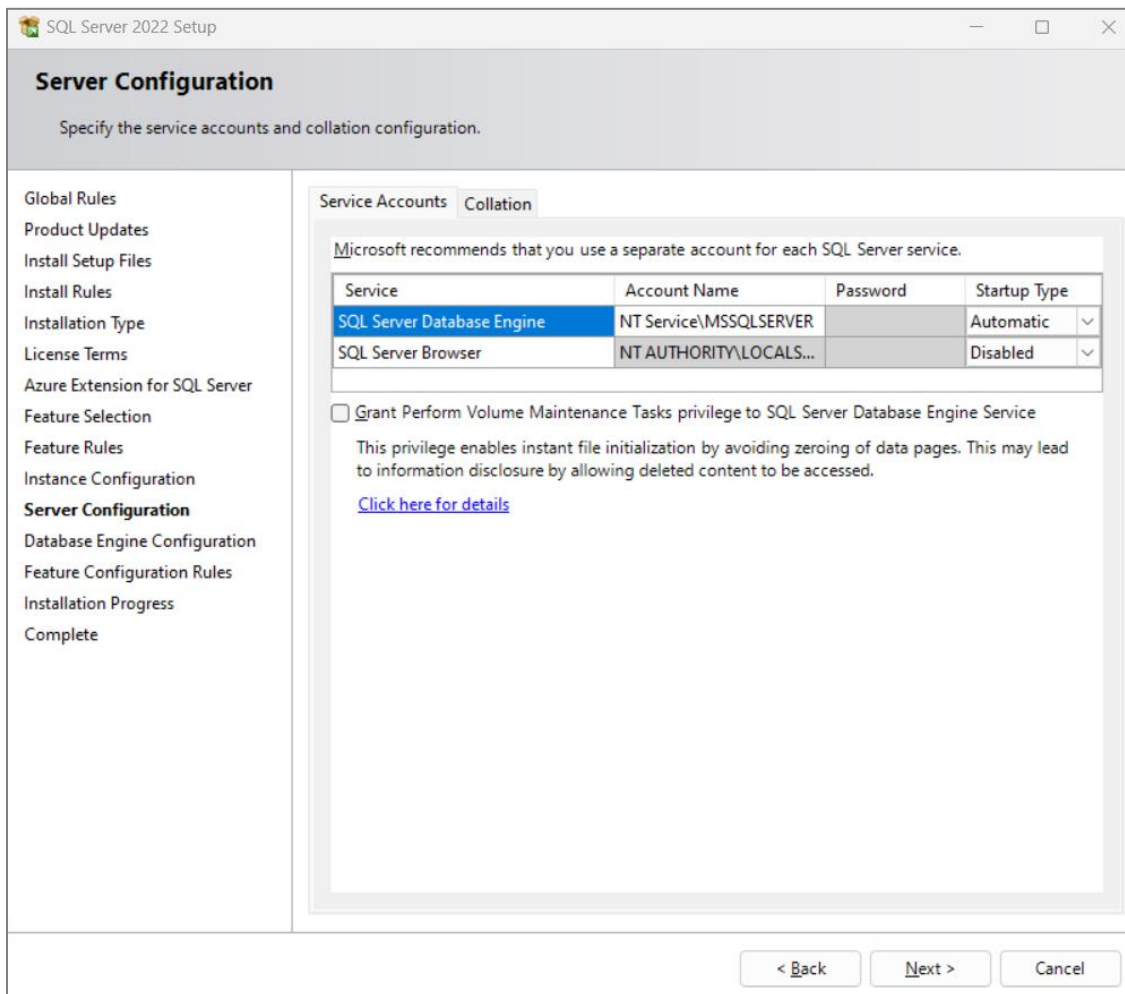
SQL Server directory: C:\Program Files\Microsoft SQL Server\MSSQL16.MSSQLSERVER

Installed instances:

Instance Name	Instance ID	Features	Edition	Version
SQLEXPRESS01	MSSQL16.SQLEXPRESS	SQLEngine	Express	16.0.1000.6
SQLEXPRESS	MSSQL10.SQLEXPRESS	SQLEngine,SQLEn...	Express	10.1.2531.0

< Back
Next >
Cancel

- **Step 14: In the Server Configuration step, change the SQL Server Database Engine service to "Automatic" and SQL Server Browser to "Disabled." Check that the accounts and startup types are specified correctly, then click "Next" to proceed.**



- **Step 15: In the Database Engine Configuration step, select the authentication mode (Windows or Mixed Mode) and identify the SQL Server administrators.**
Choose Mixed Mode, enter & confirm the password for the SQL Server system administrator (sa) account, then click "Next" to proceed.

SQL Server 2022 Setup

Database Engine Configuration

Specify Database Engine authentication security mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and Filestream settings.

- Edition
- License Terms
- Global Rules
- Microsoft Update
- Product Updates
- Install Setup Files
- Install Rules
- Azure Extension for SQL Server
- Feature Selection
- Feature Rules
- Instance Configuration
- Server Configuration
- Database Engine Configuration**
- Feature Configuration Rules
- Ready to Install
- Installation Progress
- Complete

Server Configuration | Data Directories | TempDB | MaxDOP | Memory | FILESTREAM

Specify the authentication mode and administrators for the Database Engine.

Authentication Mode

☐ Windows authentication mode

☒ Mixed Mode (SQL Server authentication and Windows authentication) 1

Specify the password for the SQL Server system administrator (sa) account.

Enter password:

Confirm password:

Specify SQL Server administrators

DESKTOP-3IIEASG\Admin (Admin) 2

SQL Server administrators have unrestricted access to the Database Engine.

3

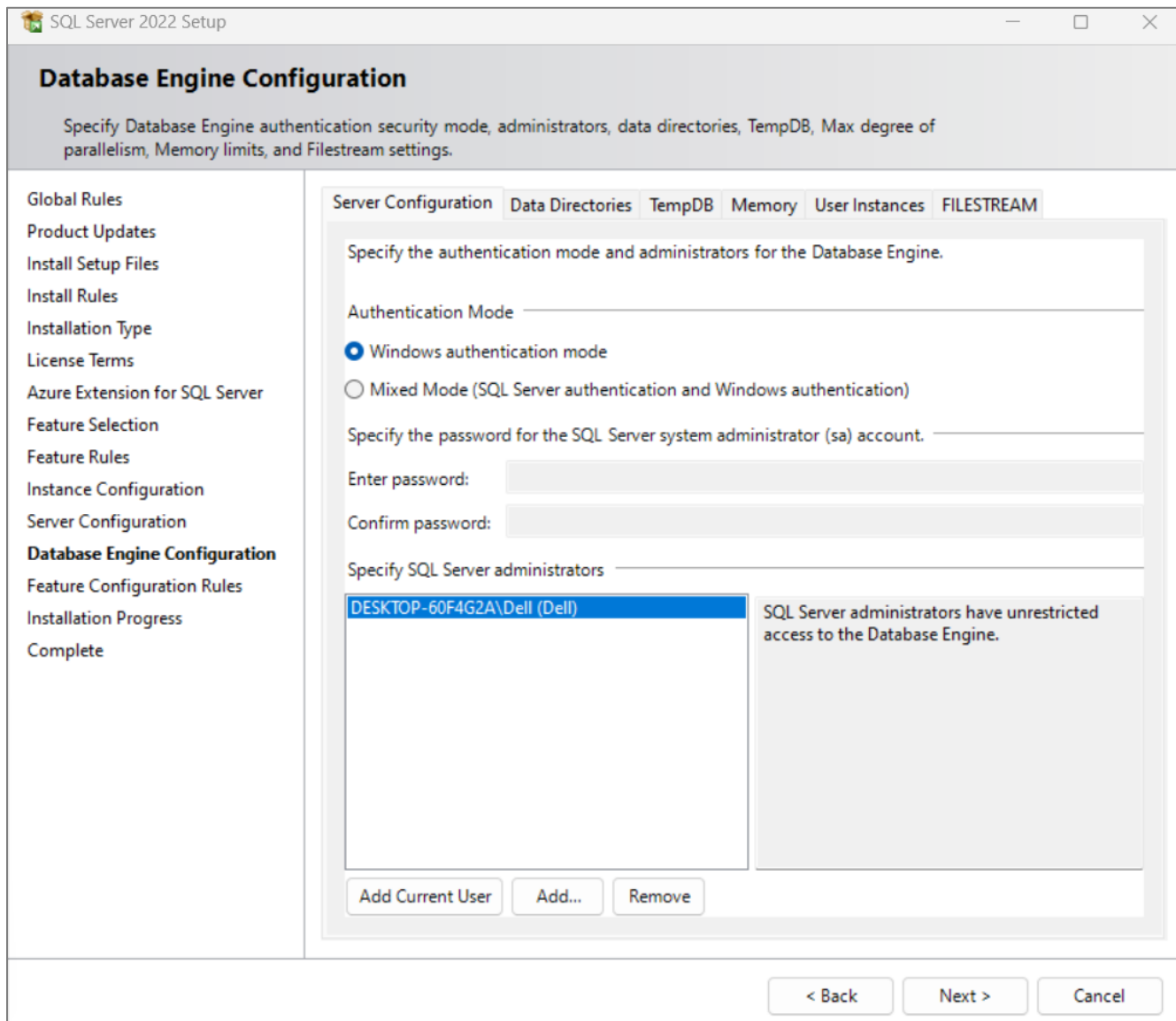
Add Current User Add... Remove

4

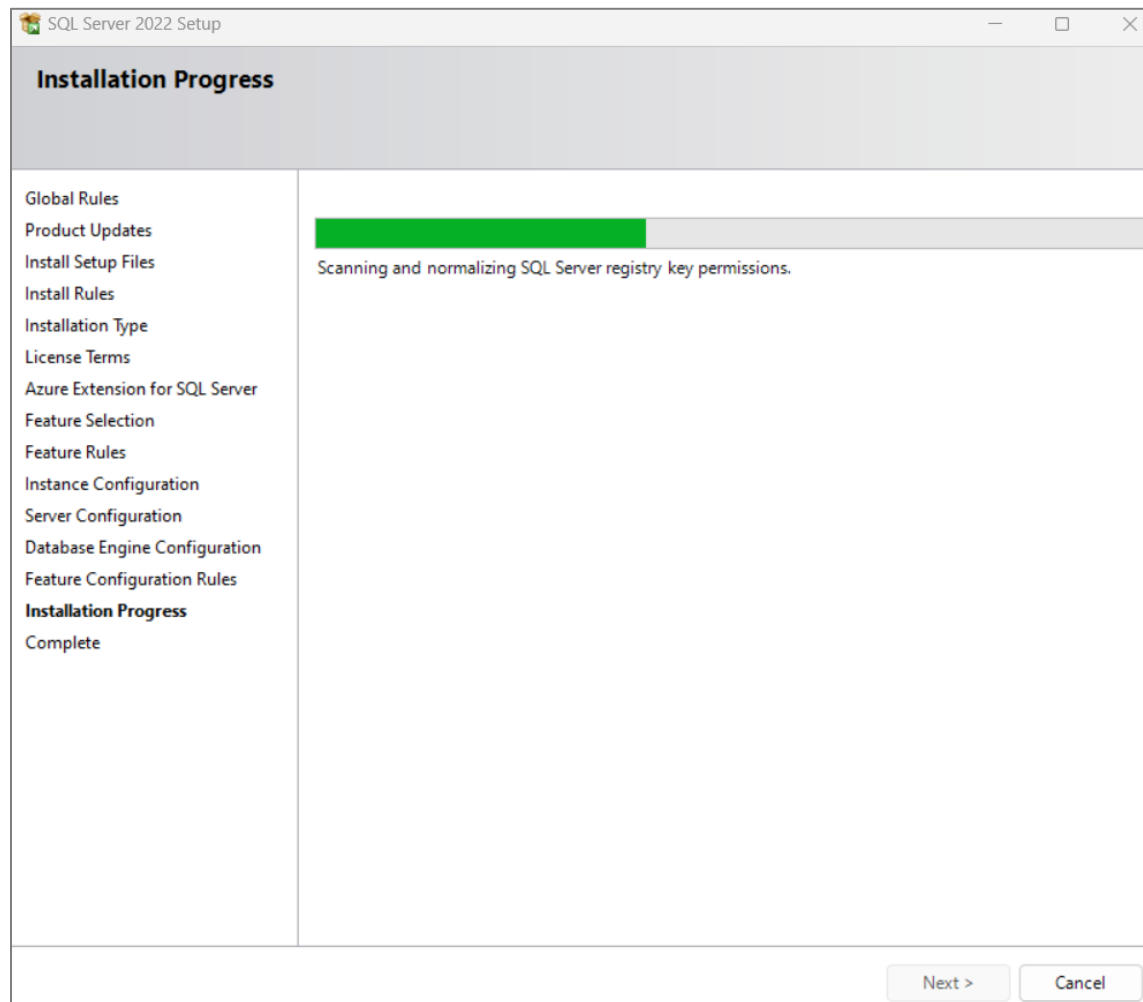
< Back Next > Cancel

Also, add Current Use as an Administrator for access.

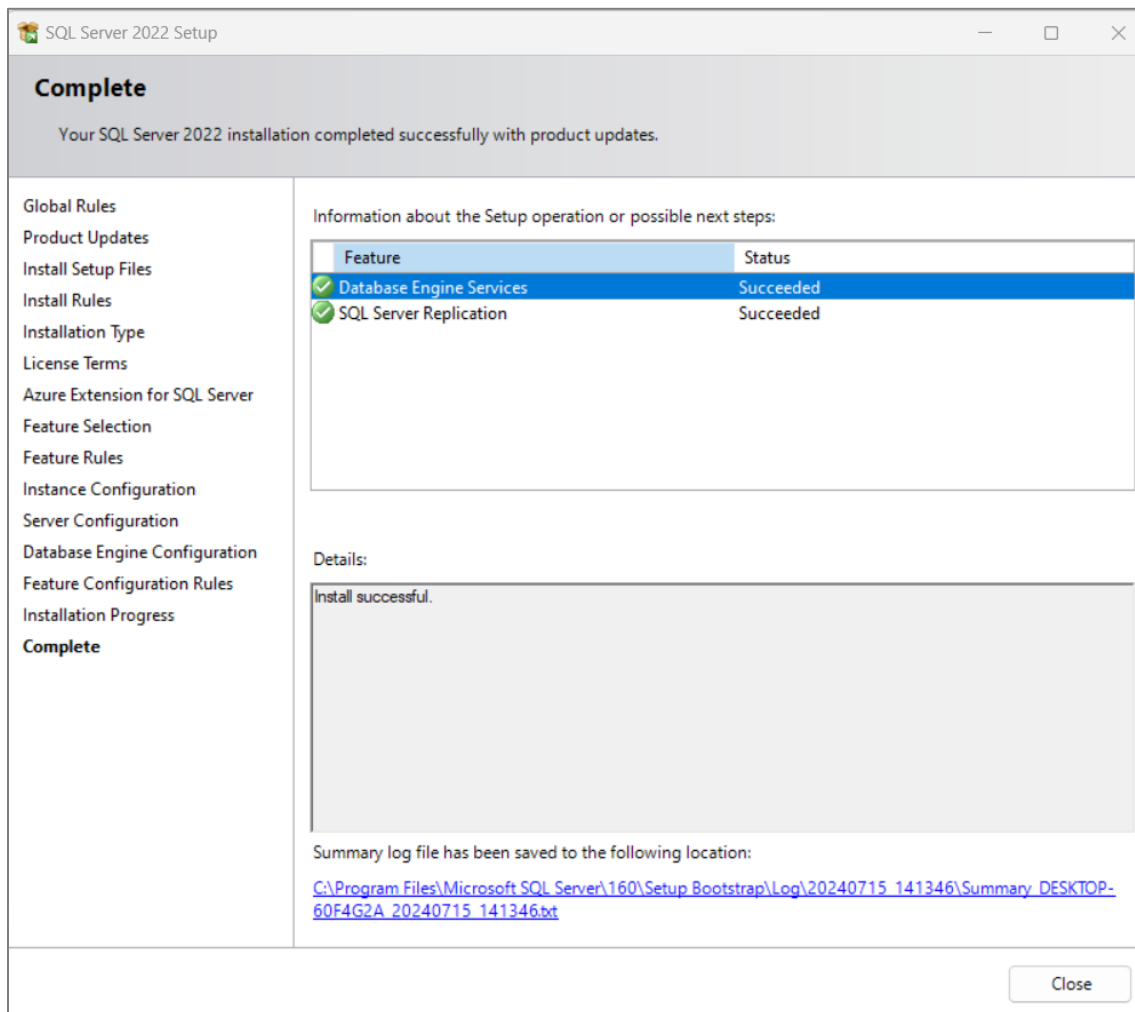
If Mixed Mode Authentication Is not working you can enable Windows Authentication mode as give below. You must enable only one mode at the time of installation.



- **Step 16: This is the "Installation Progress" stage of the SQL Server 2022 setup.**



- **Step 17: This step shows the completion of the SQL Server 2022 setup with the message "Your SQL Server 2022 installation completed successfully with product updates."** Both "Database Engine Services" and "SQL Server Replication" features have been successfully installed.

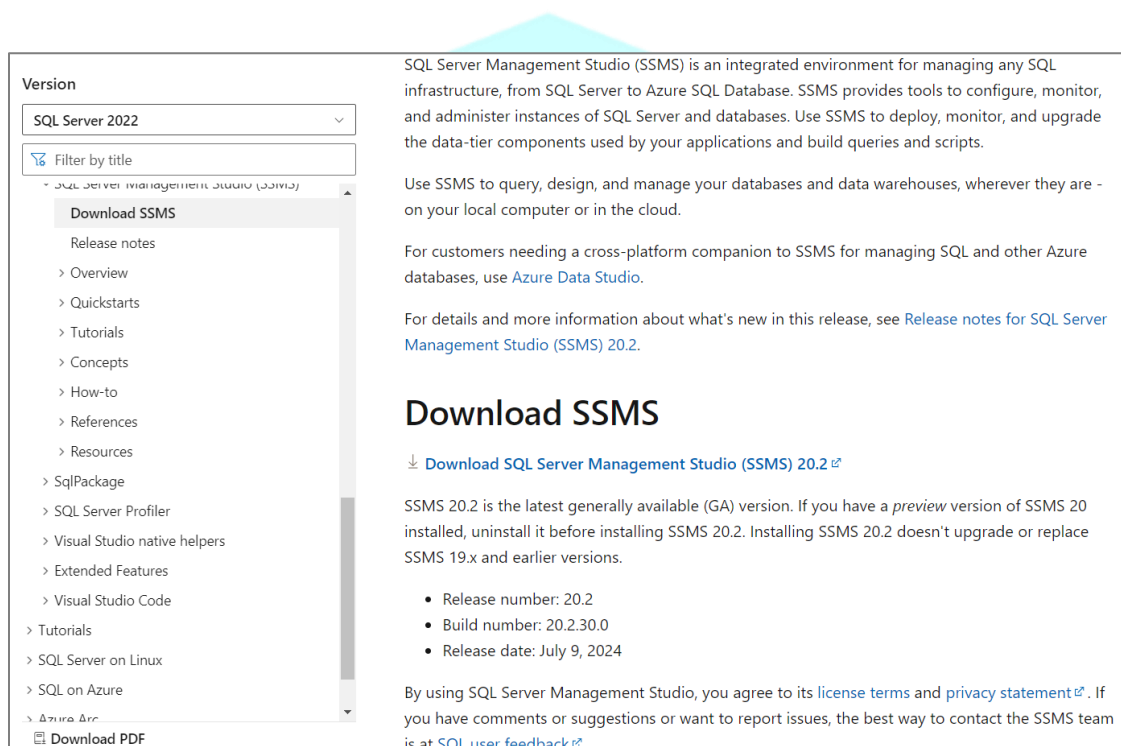


S

2.1 SQL Server Management Studio (SSMS) Installation

SQL Server Management Studio (SSMS) is an integrated environment for managing any SQL infrastructure. SSMS provides a single comprehensive utility that combines a broad group of graphical tools with many rich script editors to provide access to SQL Server for developers and database administrators of all skill levels.

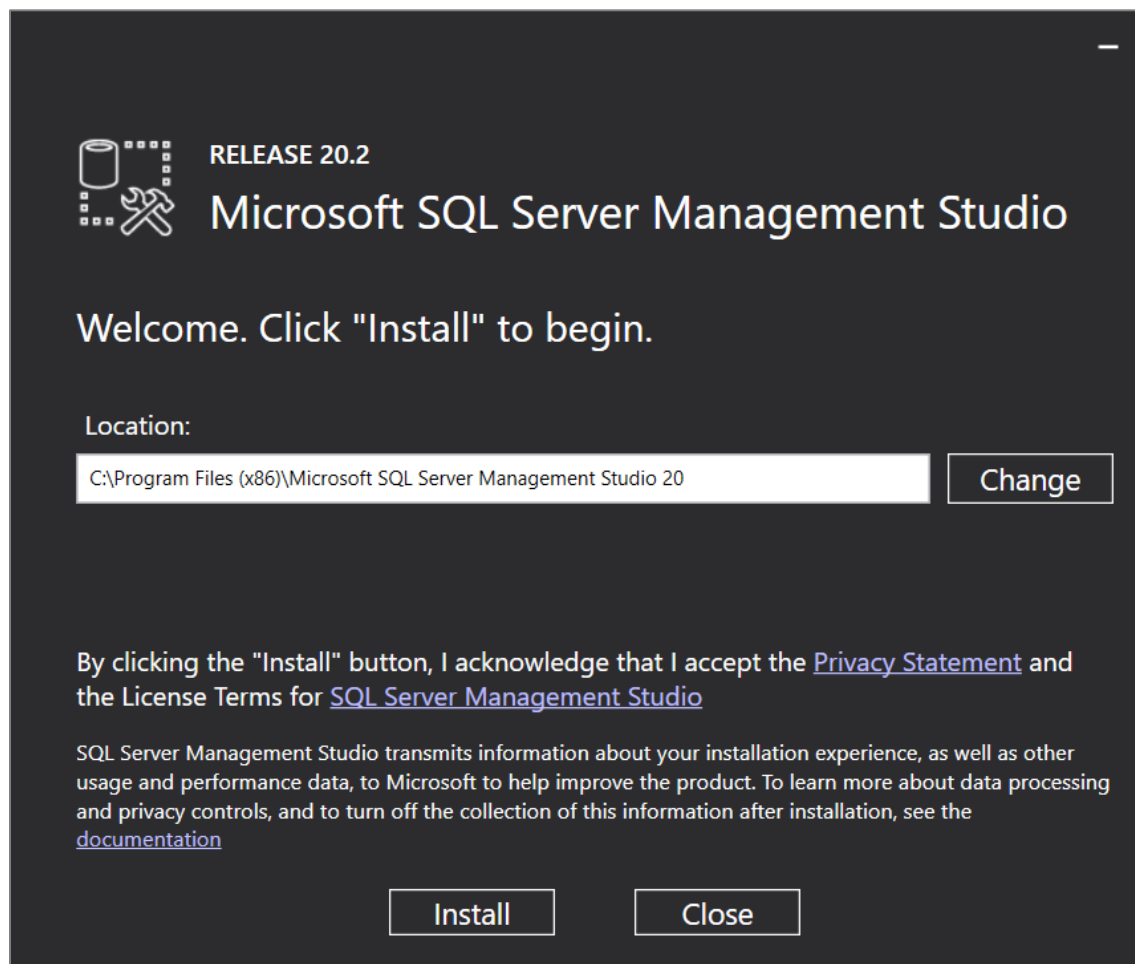
- **Step 1: Download the SQL Server Management Studio** - <https://docs.microsoft.com/en-us/sql/ssms/download-sql-server-management-studio-ssms?view=sql-server-ver15>



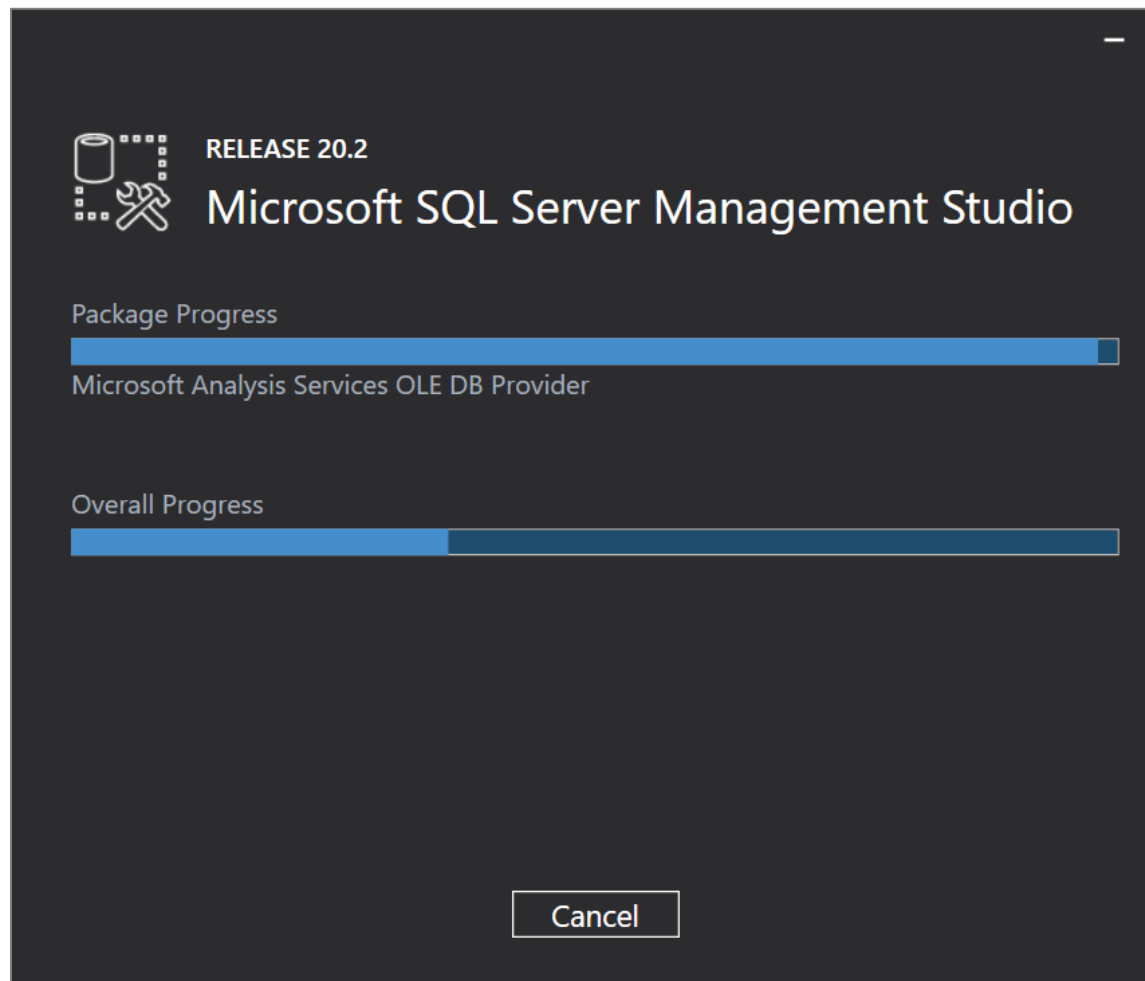
- **Step 2: Once the download is complete, go to the downloads folder in your system. Find the SSMS-Setup-ENU.exe and double-click to install SSMS.**

Name	Date modified	Type	Size
▼ Today			
 SSMS-Setup-ENU	15-07-2024 19:21	Application	4,84,561 KB

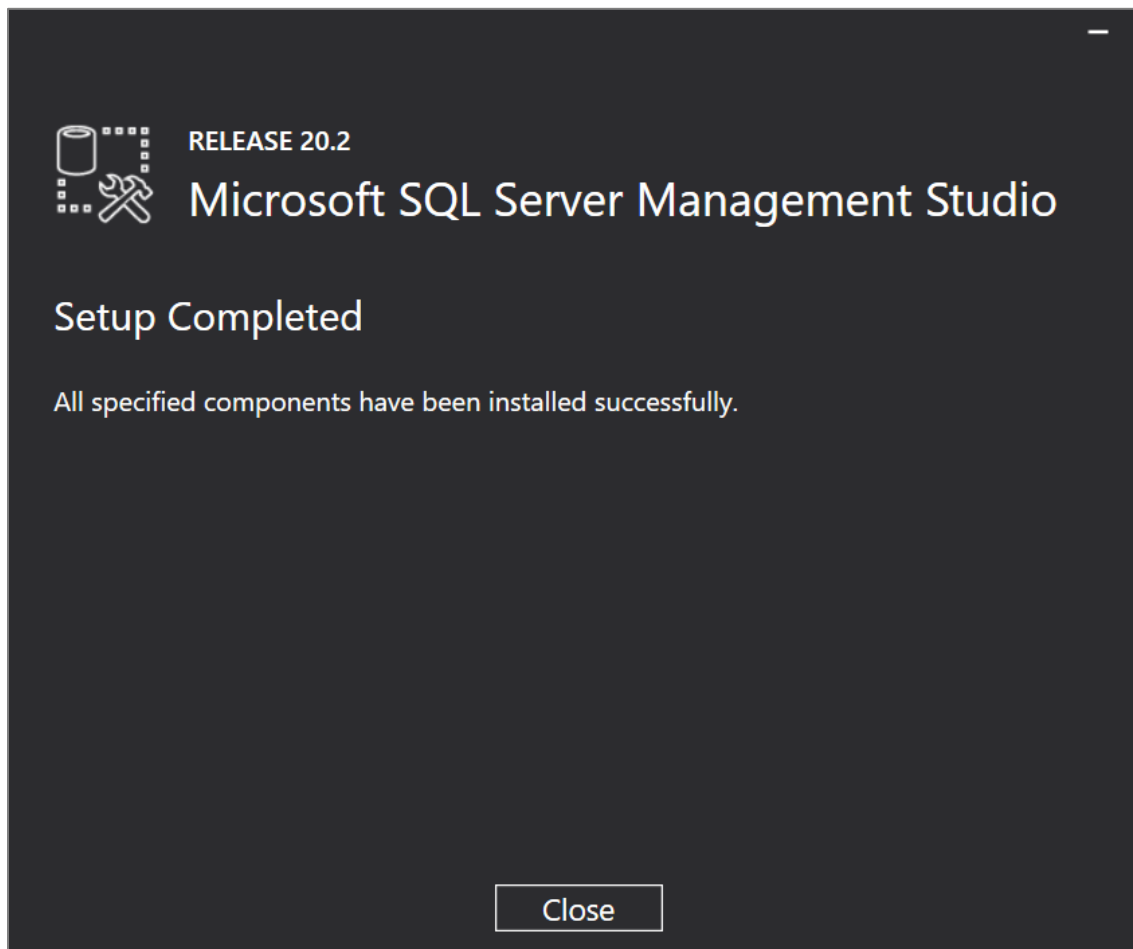
- **Step 3: An SSMS (SQL Server Management Studio) installation wizard appears.**
Click on the **Install** button to start installing SSMS.



This will take a few minutes to complete installation.

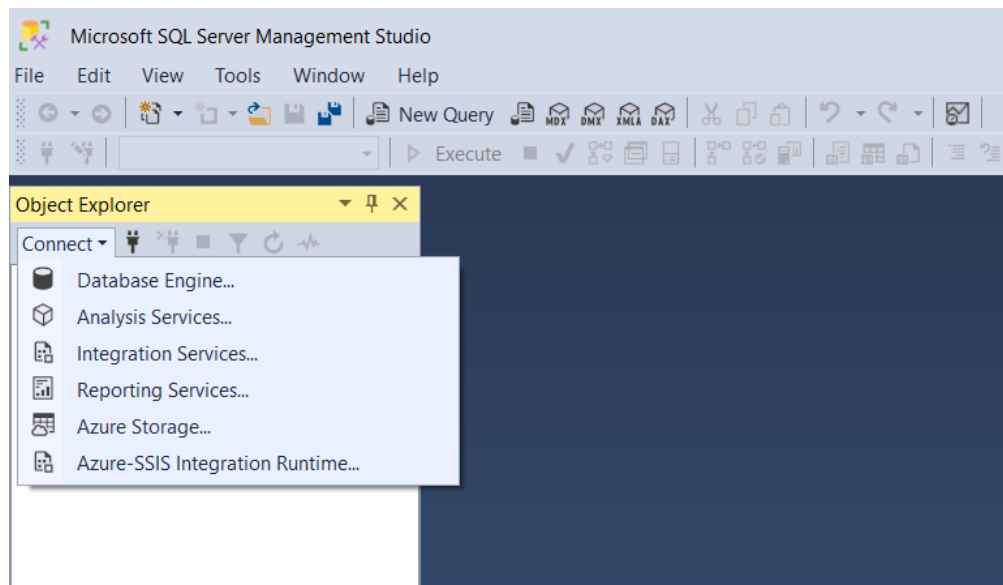


- **Step 4: After the completion of Installation, Restart the server.**
Finally, SQL Server Management Studio (SSMS) is installed on your Server.



- **Step 5: To connect to a service in SQL Server Management Studio (SSMS), first open SSMS and locate the Object Explorer on the left-hand side.**

Click the "Connect" button at the top of the Object Explorer window, and a dropdown menu will appear. From this menu, you can select the type of service you want to connect to, such as Database Engine, Analysis Services, or others.



2.2 Change SQL Server to Mixed Role Authentication:

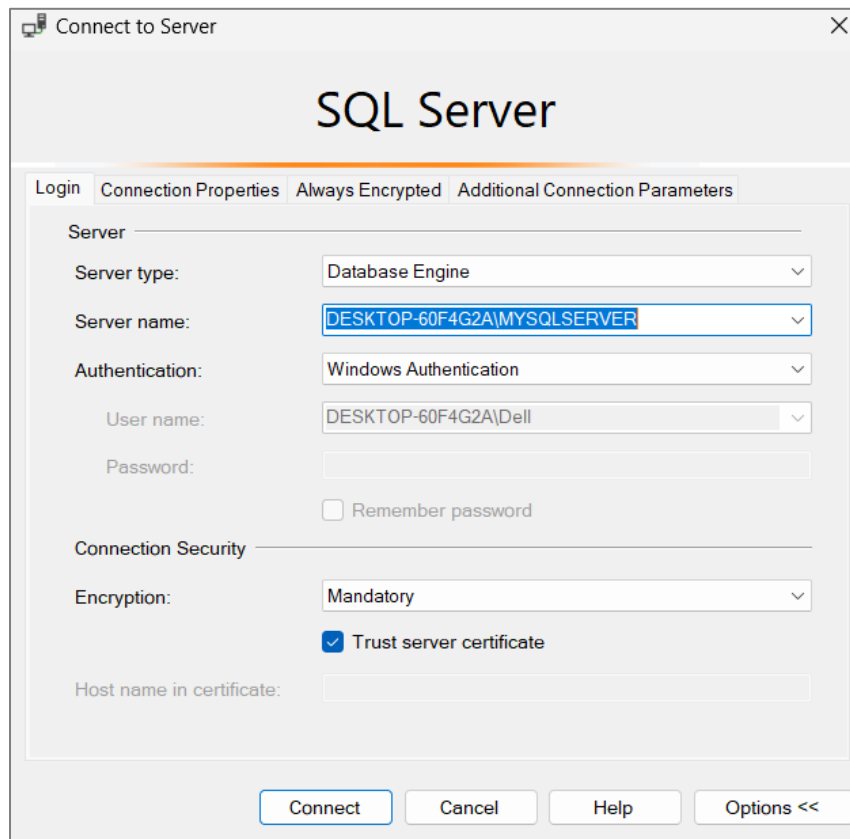
If you have configured SQL Server with Window Authentication mode and want to configure for mixed role. Then follow the given steps, otherwise skip this section.

It is considered best practice to configure SQL Server instance to allow only Windows authentication. However, some situations require that SQL Server authentication be used. There are two methods to change SQL Server to mixed mode authentication after installation.

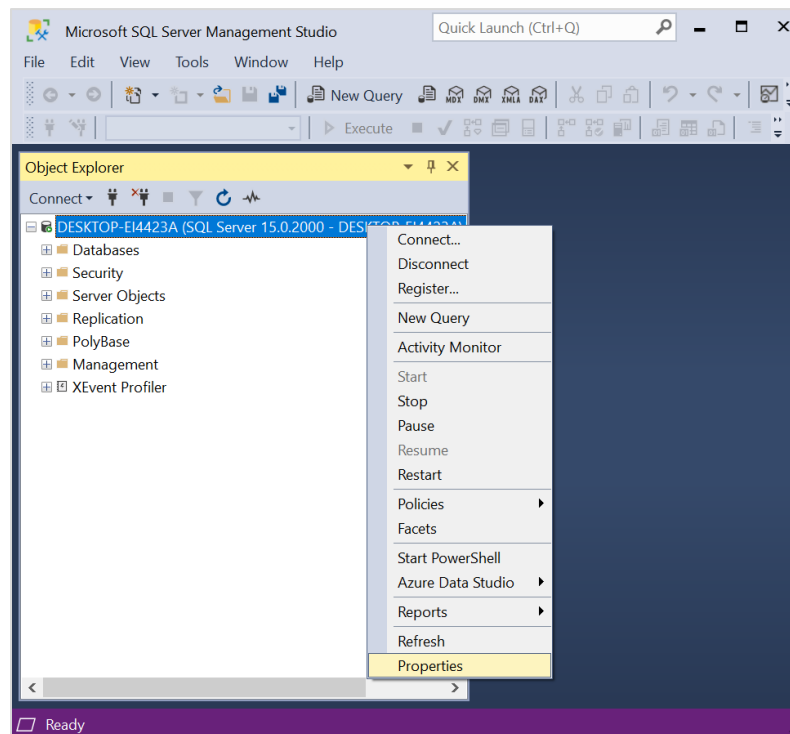
Step 1: Enable Mixed Mode Authentication in Management Studio.

If you can connect to SQL Server with Windows authentication, you can enable mixed-mode authentication easily using SQL Server Management Studio.

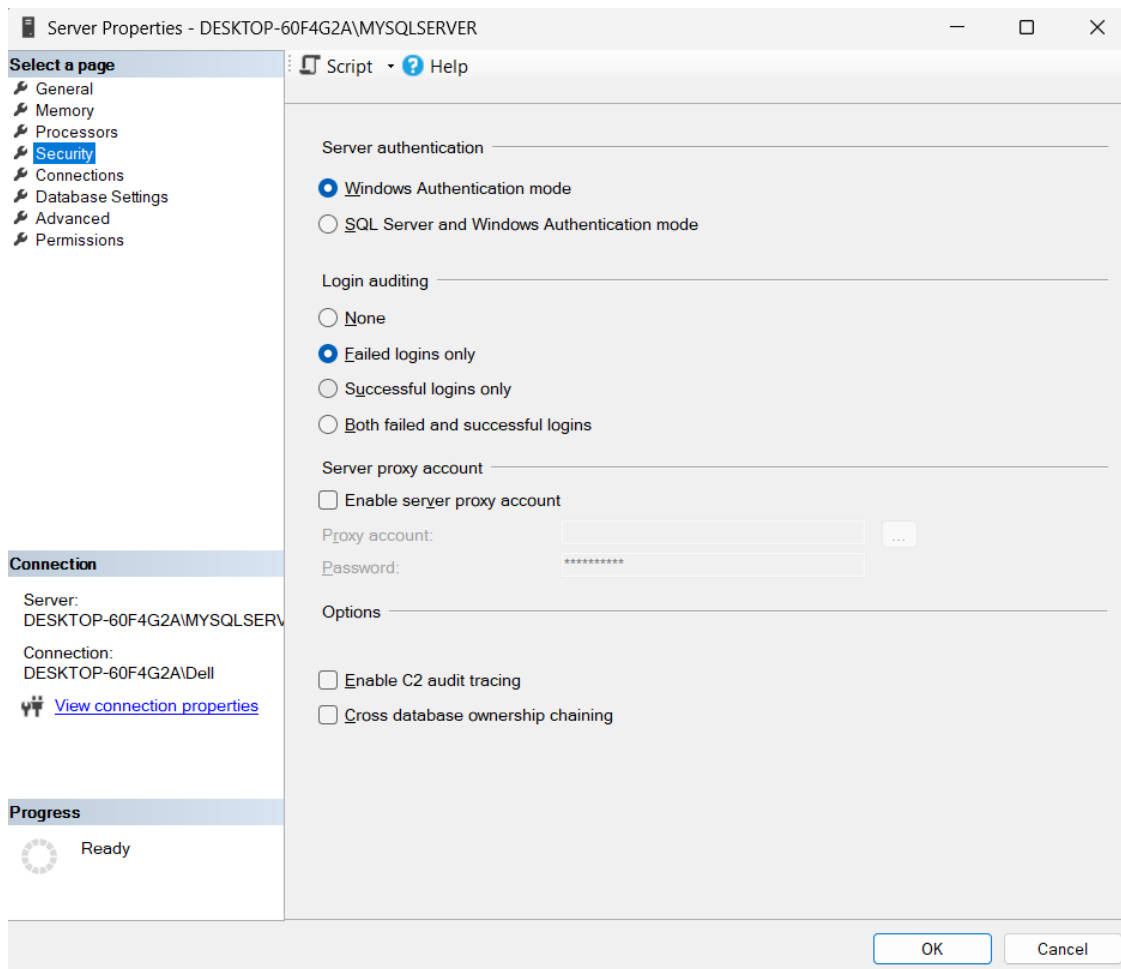
1. Login into SQL Server with SQL Server Management Studio using Windows authentication as give below.



2. In Object Explorer, right-click the server's name you wish to reconfigure and select Properties from the menu that appears.



3. This displays the Server Properties dialog box. Select the Security tab and you can then choose SQL Server and Windows Authentication mode (also known as mixed mode authentication).



4. Click OK to save the changes and restart the SQL Server service.

Step 2: To Connect with Windows Authentication Mode:

1. Open SQL Server Management Studio (SSMS).
2. Go to the "File" menu.
3. Click on "Connect Object Explorer".
4. In the "Connect to Server" dialog box, set "Server type" to "Database Engine".
5. Enter the "Server name" (e.g., DESKTOP-60F4G2A\MYSQLSERVER). If you don't know it, you can browse for more.
6. Ensure "Authentication" is set to "Windows Authentication".
7. Verify the "User name" is correct (e.g., DESKTOP-60F4G2A\Dell).
8. Under "Connection Security", set "Encryption" to "Mandatory".
9. Check the box for "Trust server certificate".

10. Click on the "**Connect**" button to establish the connection.

Connect to Server

SQL Server

Login | Connection Properties | Always Encrypted | Additional Connection Parameters

Server

Server type: Database Engine

Server name: DESKTOP-60F4G2A\MYSQLSERVER

Authentication: Windows Authentication

User name: DESKTOP-60F4G2A\DeI

Password:

☐ Remember password

Connection Security

Encryption: Mandatory

☒ Trust server certificate

Host name in certificate:

Connect Cancel Help Options <<

Step 3: To Connect with SQL Server Authentication Mode:

1. Open SQL Server Management Studio (SSMS).
2. Go to the "**File**" menu.
3. Click on "**Connect Object Explorer**".
4. In the "Connect to Server" dialog box, set "Server type" to "Database Engine".
5. Enter the "Server name" (e.g., DESKTOP-60F4G2A\MYSQLSERVER). If you don't know it, you can browse for more.
6. From the "Authentication" drop-down menu, select "**SQL Server Authentication**".
7. The Login name will default to "sa" (System Admin).

8. Enter the "Password" that you set during the installation of the SQL Server.
9. Under "Connection Security", set "Encryption" to "Mandatory".
10. Check the box for "Trust server certificate".
11. Click on the "**Connect**" button to establish the connection.

Connect to Server

SQL Server

Login | Connection Properties | Always Encrypted | Additional Connection Parameters

Server

Server type: Database Engine

Server name: DESKTOP-60F4G2A\MYSQLSERVER

Authentication: SQL Server Authentication

Login: sa

Password: *****

☐ Remember password

Connection Security

Encryption: Mandatory

☒ Trust server certificate

Host name in certificate:

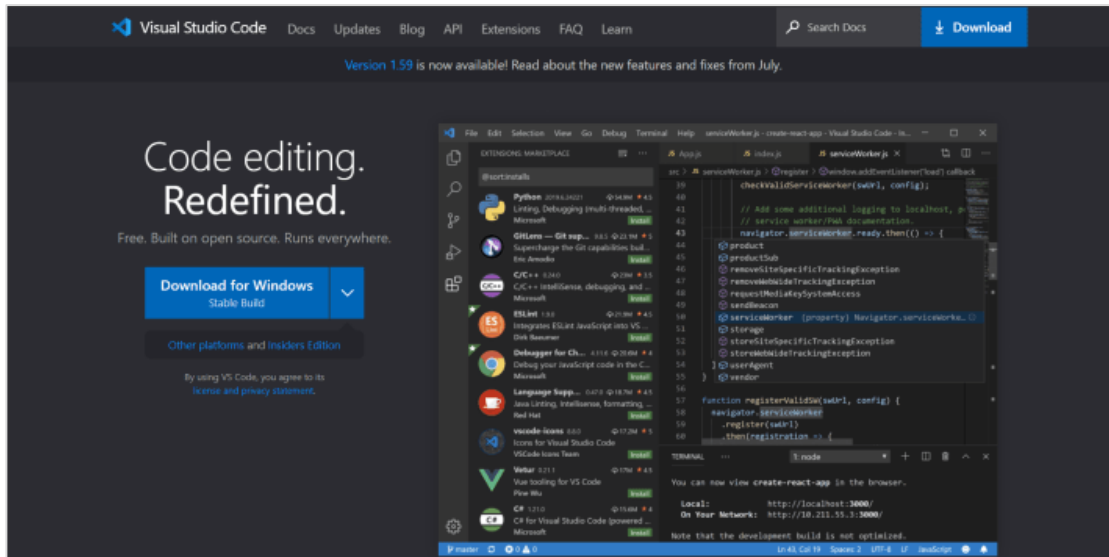
Connect Cancel Help Options <<

3. Visual Studio Code Installation

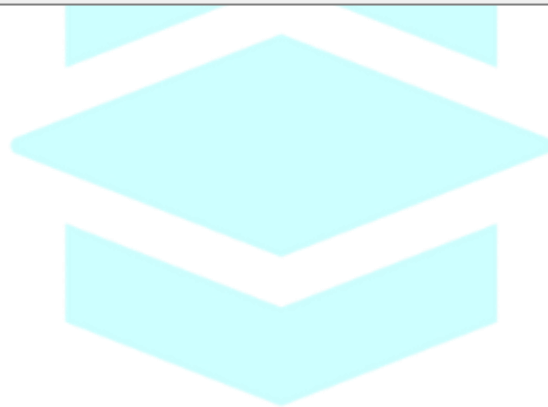
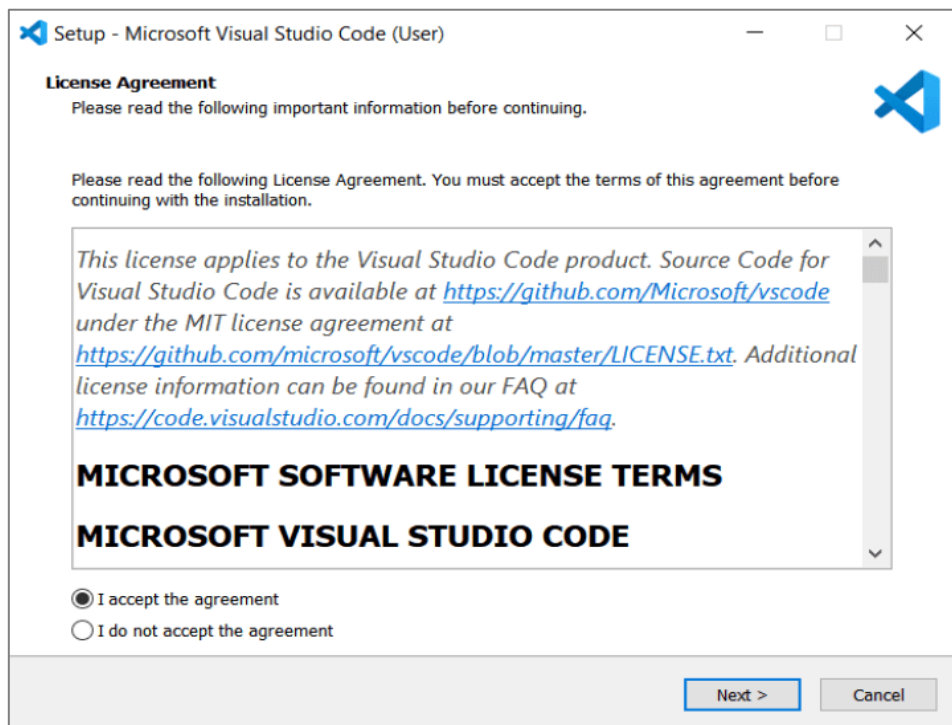
VS Code is an open-source code editor developed by Microsoft for Windows, Linux, and Mac OS platforms. Here we discuss the steps to install Visual Studio Code or VS code on Windows. It has

built-in support for TypeScript, JavaScript, Node.js, and almost all programming language/library/framework-based plugins are available in VS studio right now.

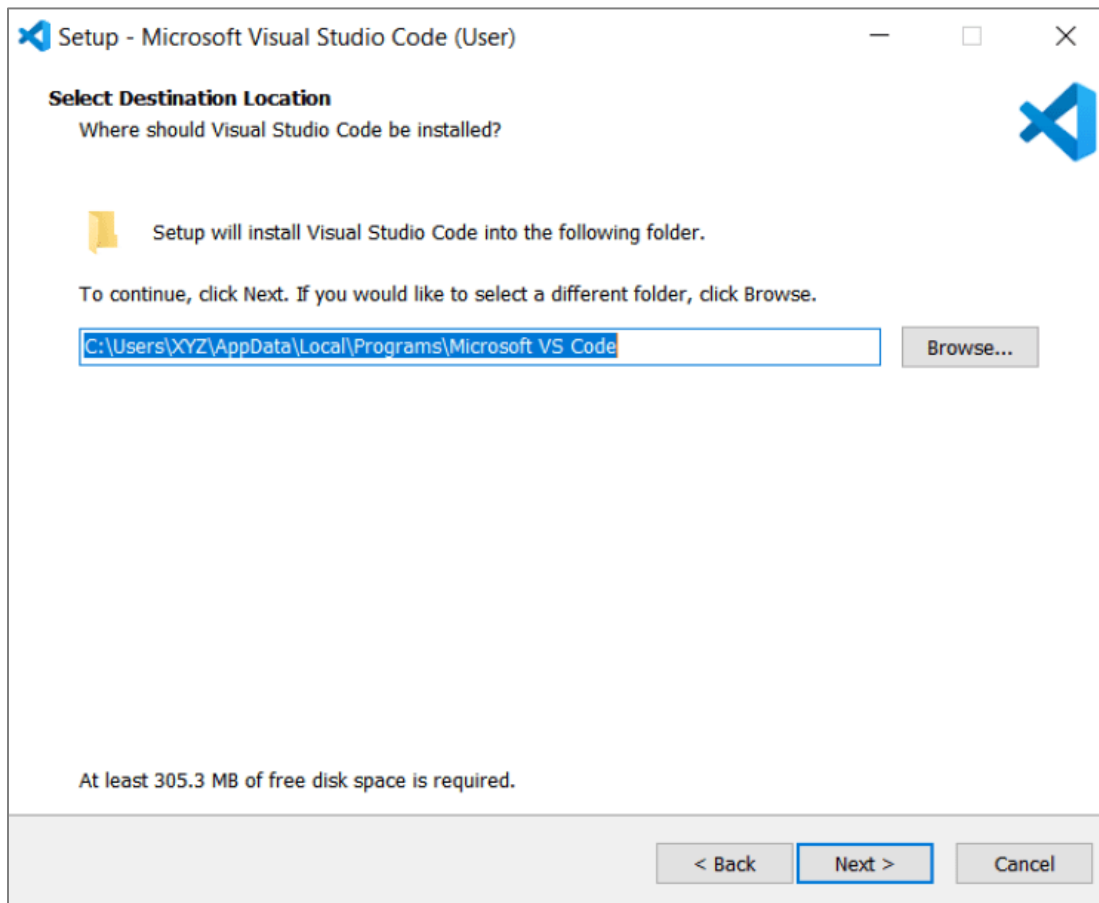
- **Step 1:** Go to <https://code.visualstudio.com/> and download the Setup file according to your Operating System.



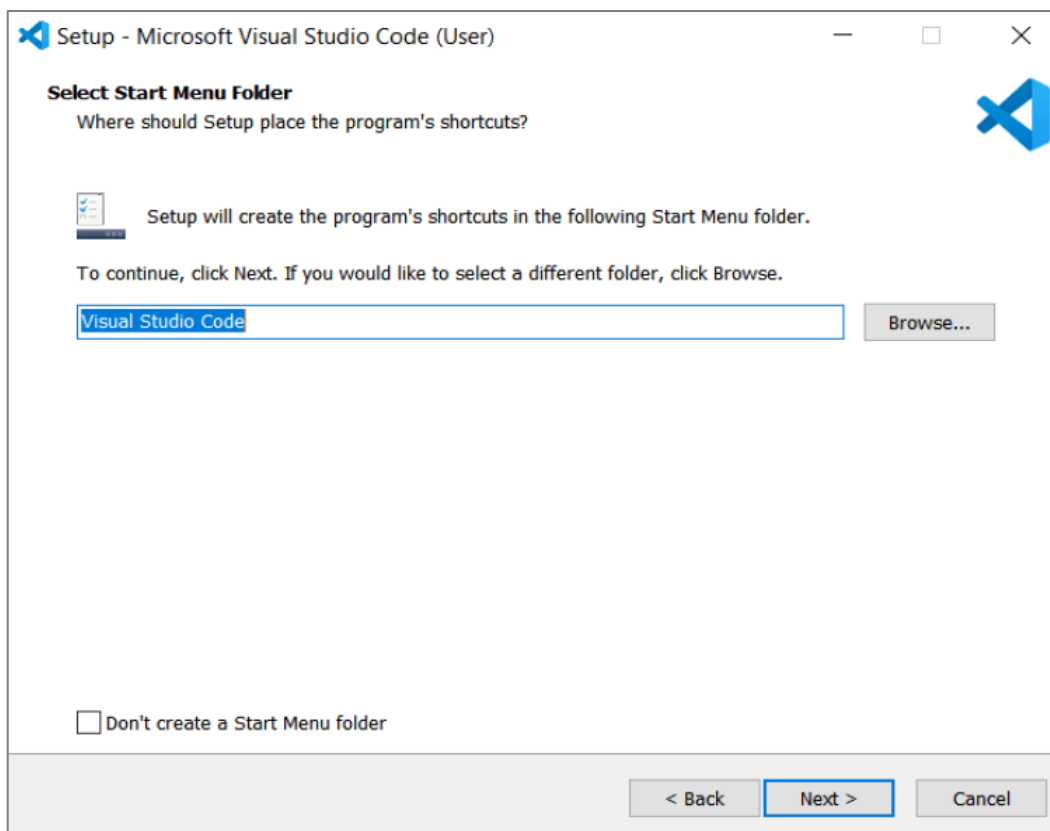
- **Step 2:** After Downloading the file from the link, open the installer.
- **Step 3:** Accept the license agreement and click on Next.



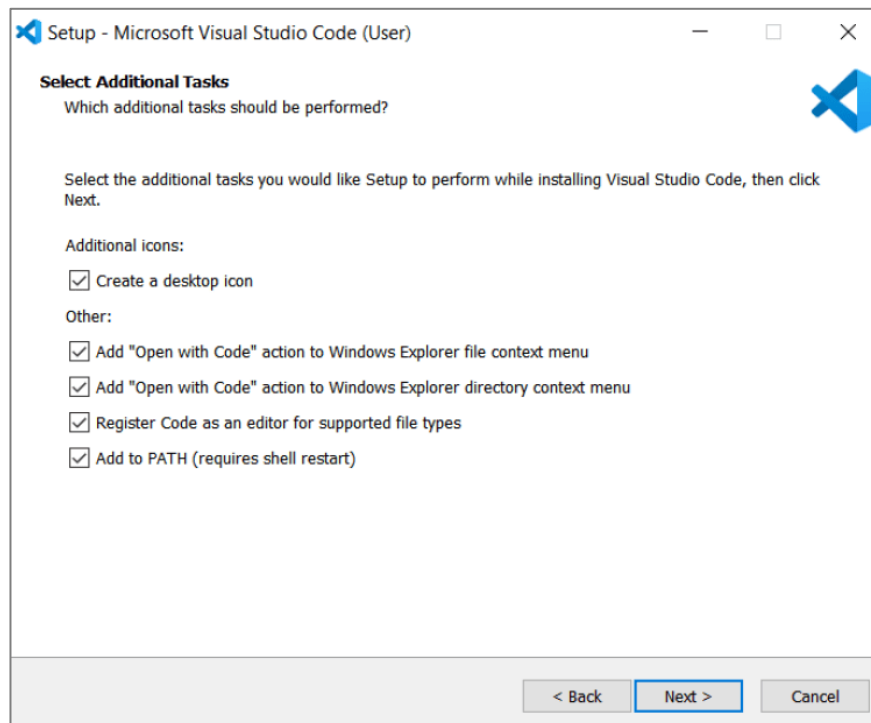
- **Step 4:** You can Select the destination where you need to install, by default it will be installed in the folder given in the below image.



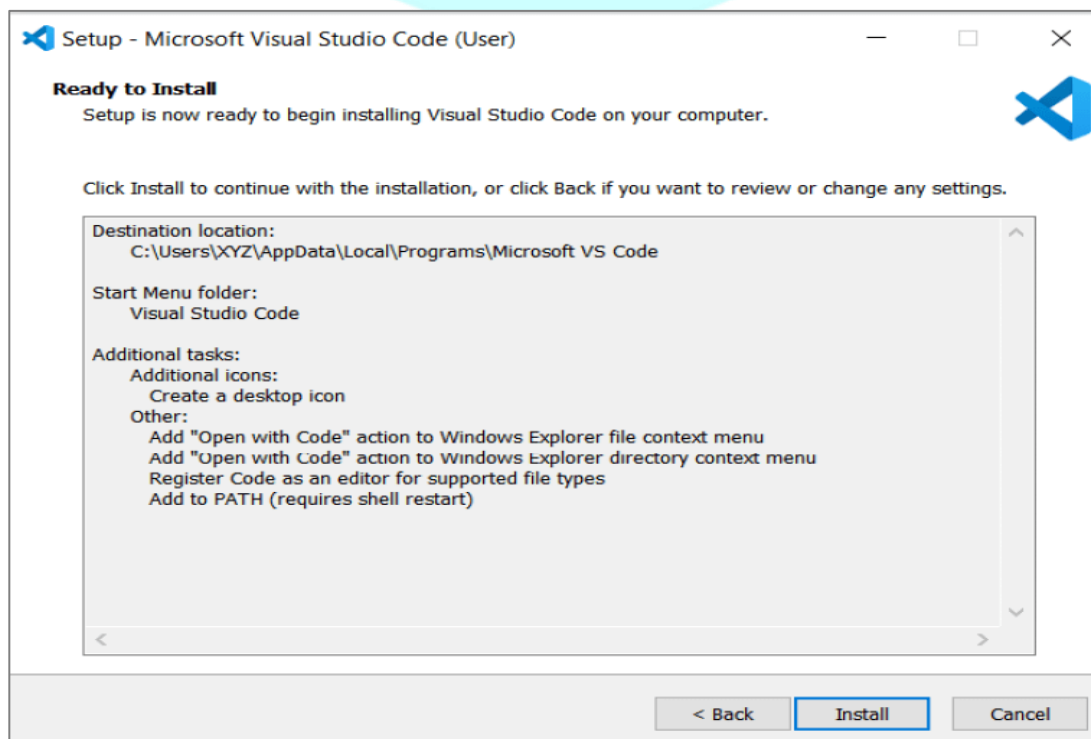
- **Step 5: Click on Next and It will create folder in start menu.**



Step 6: Click all the checkboxes and it will be helpful and it will also add to the PATH.



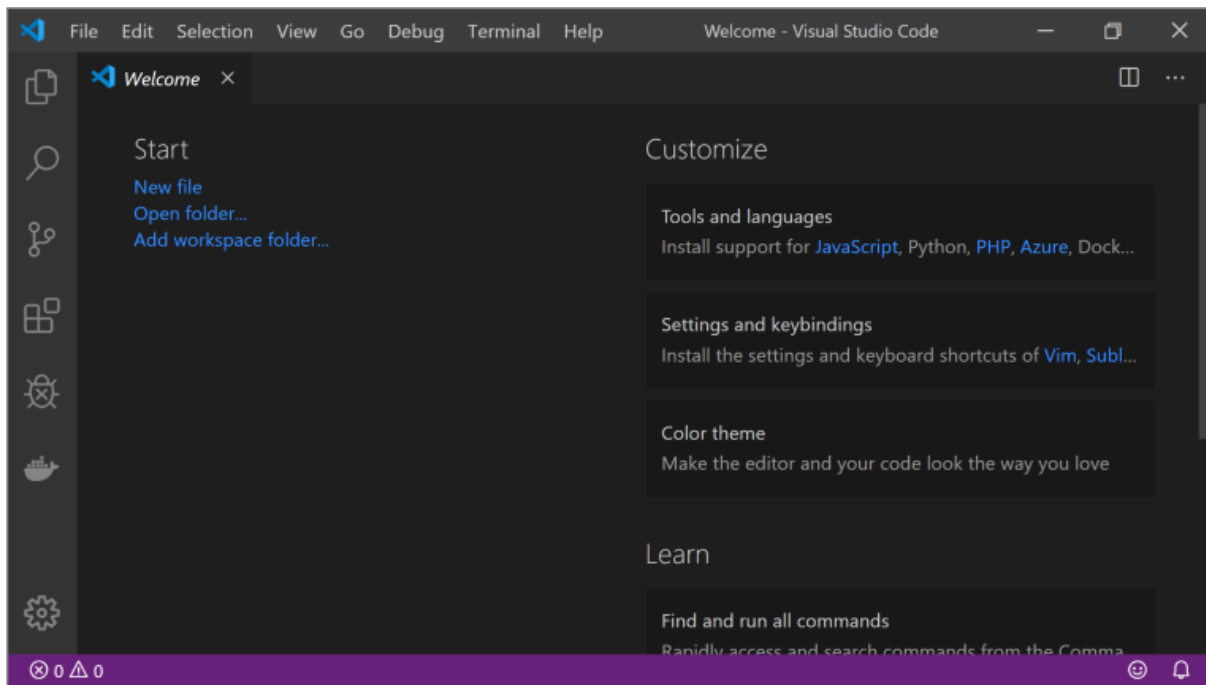
● Step 7: Review Settings



Now it will show all your settings there to install VS Code.

- **Step 8: Launch VS Code**

Then you will see the Welcome Screen of VS Code as shown below.



Visual Studio Code has successfully installed. Let the programming fun begin!

4. WSL2 Installation

This document will help you to install the WSL2 on your windows operating system. For the latest steps help, please do refer here: [Manual installation steps for WSL | Microsoft Learn](#)

- **Step 1 - Enable the Windows Subsystem for Linux**

You must first enable the "Windows Subsystem for Linux" optional feature before installing any Linux distributions on Windows.

Open PowerShell as Administrator (Start menu > PowerShell > right-click > Run as Administrator) and enter this command:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
```

- **Step 2 - Check requirements for running WSL 2**

To update to WSL 2, you must be running Windows 10 or Windows 11

- For x64 systems: Version 1903 or later, with Build 18362.1049 or later.
- For ARM64 systems: Version 2004 or later, with Build 19041 or later.

- **Step 3 - Enable Virtual Machine feature**

Before installing WSL 2, you must enable the Virtual Machine Platform optional feature. Your machine will require virtualization capabilities to use this feature.

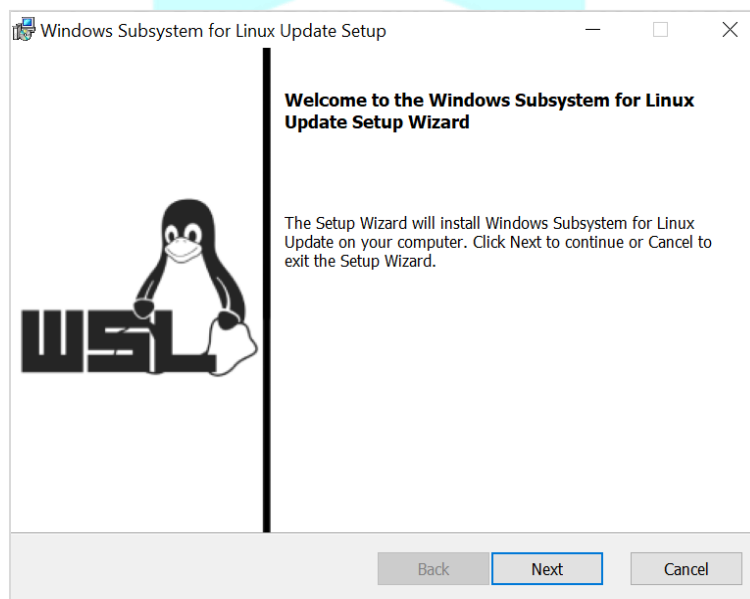
Open PowerShell as Administrator and run:

```
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

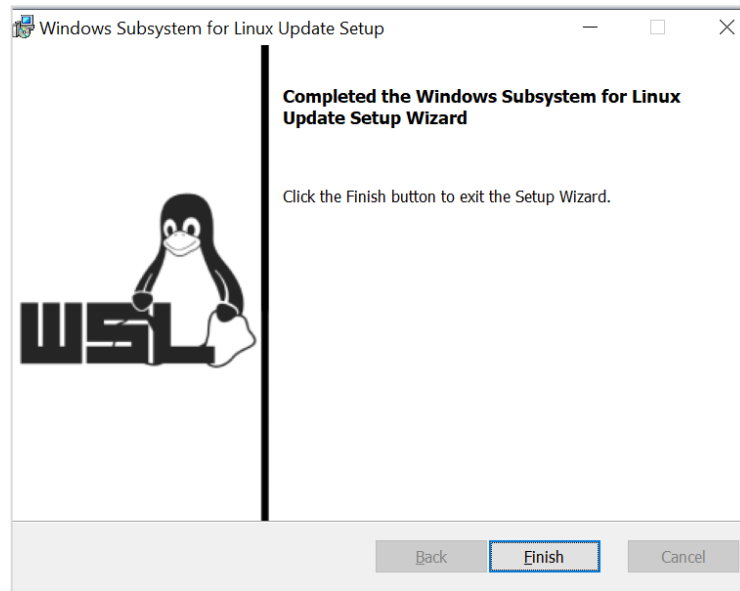
- **Step 4 - Download the Linux kernel update package**

Download the latest package: [WSL2 Linux kernel update package for x64 machines](#)

Now install it by following the steps.



Once the installation is completed, Click on Finish.



- **Step 5 - Set WSL 2 as your default version**

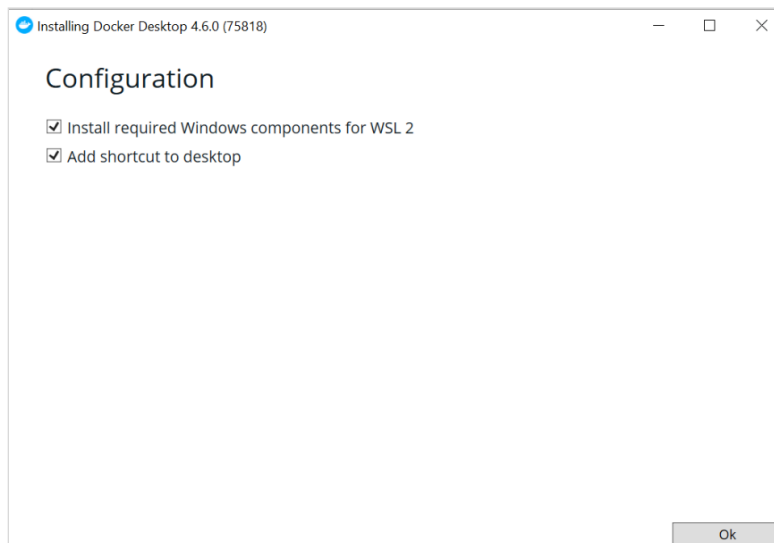
Open PowerShell and run this command to set WSL 2 as the default version when installing a new Linux distribution:

```
wsl --set-default-version 2
```

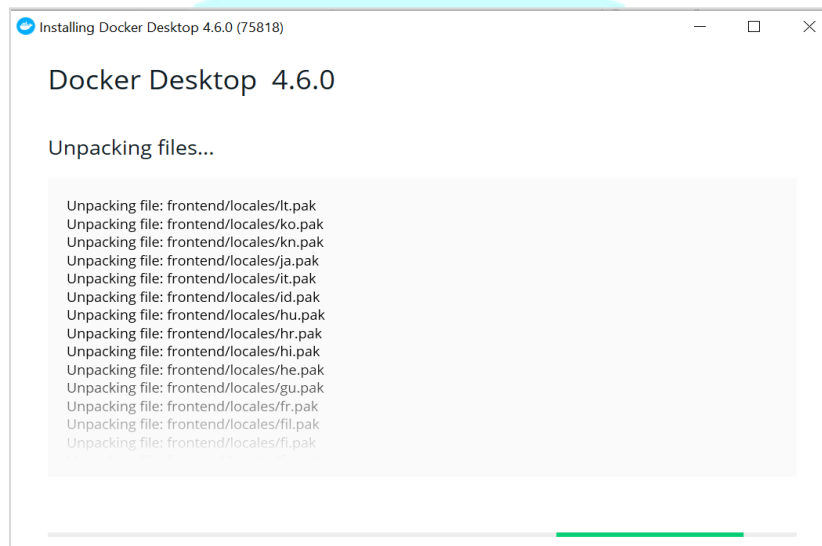
5. Docker Desktop Installation

This document will help you to install the Docker Desktop on your windows operating system.

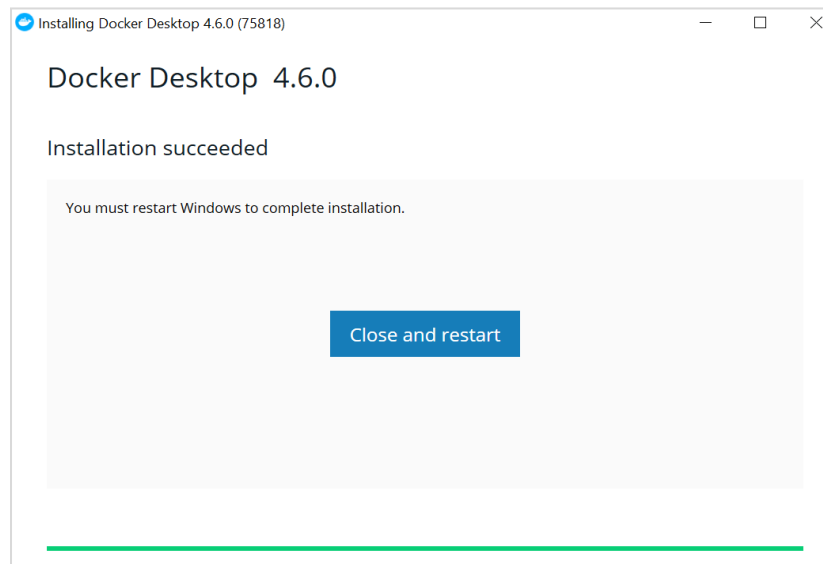
- **Step 1: Download the Docker for Windows. Go to the path/folder where the file has been downloaded and Open Installer.**
- **Step 2: Open the Installer file, now you will see Downloading Message.**
- **Step 3: Tick both the configuration options and click on Next.**



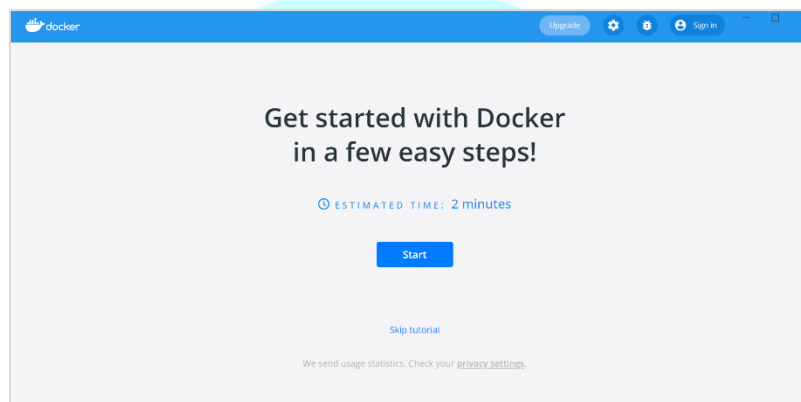
Now it will start Installing the Docker and it will take a couple of minutes.



And within a few minutes, you will get Installation Succeeded Message.



After Complete Installation, Docker gets started!



- **Step 4: To verify whether Docker is installed properly or not – Go to Command Prompt. Type the following command:**

docker version

If you can see Client-server Details, then it is Installed Completely.

```
C:\Users\Miles pc>docker version
Client:
 Cloud integration: v1.0.22
 Version:          20.10.13
 API version:      1.41
 Go version:       go1.16.15
 Git commit:       a224086
 Built:            Thu Mar 10 14:13:04 2022
 OS/Arch:          windows/amd64
 Context:          default
 Experimental:     true

Server: Docker Desktop 4.6.0 (75818)
Engine:
 Version:          20.10.13
 API version:      1.41 (minimum version 1.12)
 Go version:       go1.16.15
 Git commit:       906f57f
 Built:            Thu Mar 10 14:06:05 2022
 OS/Arch:          linux/amd64
 Experimental:     false
containerd:
 Version:          1.5.10
 GitCommit:        2a1d4dbdb2a1030dc5b01e96fb110a9d9f150ecc
runc:
 Version:          1.0.3
 GitCommit:        v1.0.3-0-gf46b6ba
docker-init:
 Version:          0.19.0
```

6. Kubernetes Setup Using Docker Desktop

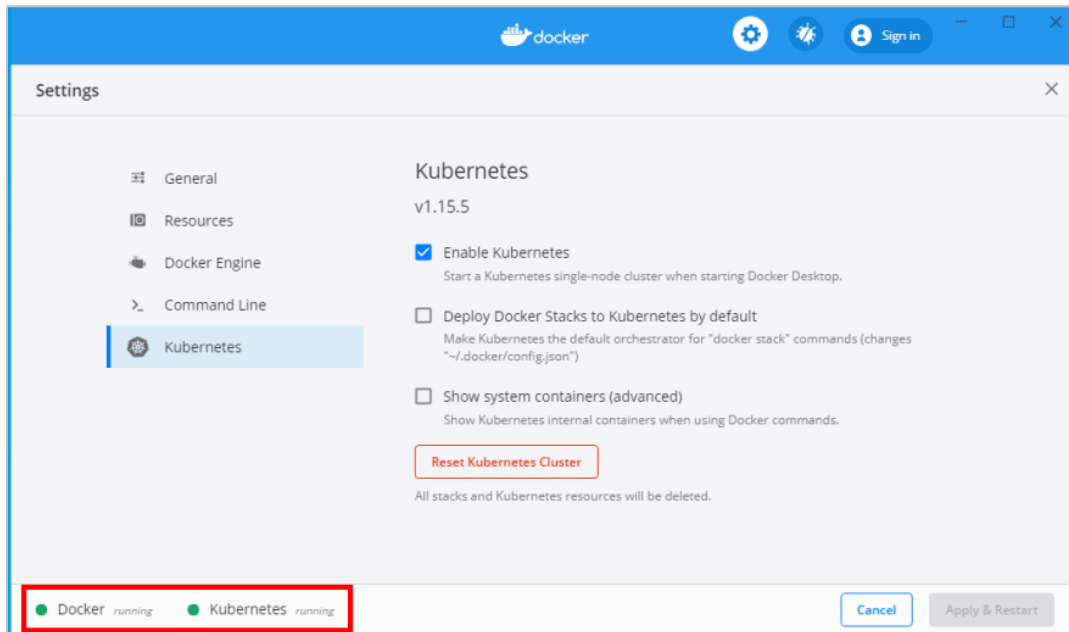
Docker Desktop simplifies the developing applications for Kubernetes. It provides a simple Kubernetes setup experience by hiding the complexity of the installation and networking details. Developers can focus entirely on their work rather than dealing with the Kubernetes setup details.

- **Step 1: Install Kubernetes on Windows 10.**

1. Right-click the Docker tray icon.
2. Click "Settings".
3. On the left panel click "Kubernetes".
4. Check Enable Kubernetes and click "Apply".

Wait until the 'Installation complete!' popup message is shown up.

If both services (Docker & Kubernetes) are running successfully without errors, both icons at the bottom left will go green.



- **Step 2: Install Kubernetes Dashboard.**

Kubernetes Dashboard is not installed by default. Run the following commands to deploy & enable the Kubernetes Dashboard using the following commands.

1. **Download the yaml configuration file**
from <https://raw.githubusercontent.com/kubernetes/dashboard/v2.0.0-rc7/aio/deploy/recommended.yaml>
2. **Deploy it using kubectl apply -f .\recommended.yaml**
3. **Verify that it's running using the following command: kubectl.exe get -f .\recommended.yaml.txt**

```
PS C:\Users\user\Desktop> kubectl.exe get -f .\recommended.yaml.txt

NAME                                STATUS    AGE
namespace/kubernetes-dashboard      Active    2m10s

NAME                                SECRETS    AGE
serviceaccount/kubernetes-dashboard  1          2m10s

NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/kubernetes-dashboard        ClusterIP    10.97.3.127    <none>         443/TCP    2m9s
```

- **Step 3: Access the Dashboard.**

There are two ways to access the dashboard with tokens:

the first one is using the default token that was created during the installation of Kubernetes and the second (more secure) way is by creating users, giving them permissions, and then get the generated token. we will use the first way.

1. Run the following command PowerShell (not cmd) ((kubectl -n Kube-system describe secret default | Select-String "token:") -split " ")[1]
2. Copy the generated token
3. Run kubectl proxy.
4. Open the following link on your browser: <http://localhost:8001/api/v1/namespaces/kubernetes-dashboard/services/https:kubernetes-dashboard:/proxy/>
5. Select Token & paste the generated token
6. Sign In.

Step 4: Finally!

If everything is configured properly, you'll be able to see the dashboard and your cloud resources. From there you can do almost all the "hard" work without dealing with the CLI every time.

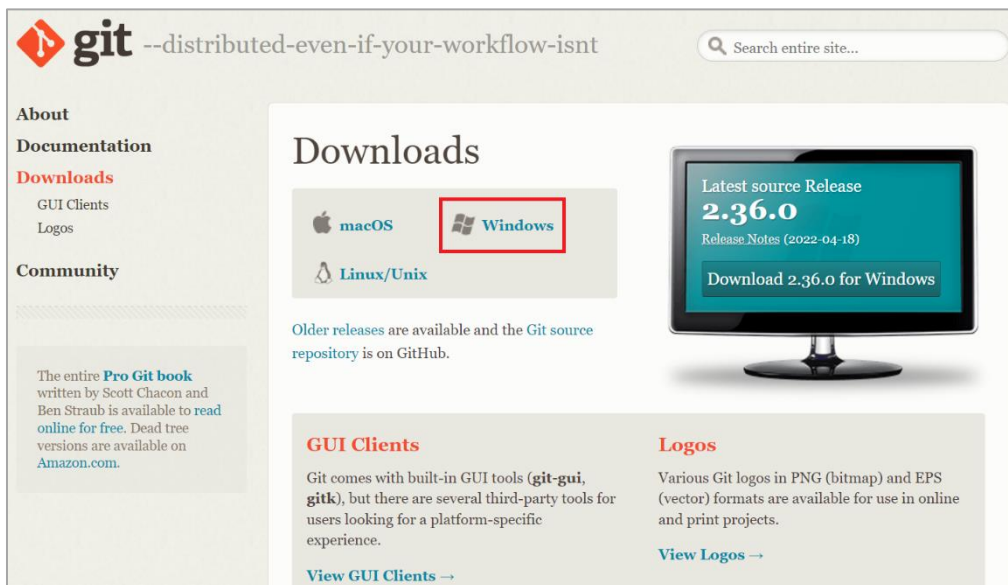
7. Git Installation

Git is a widely used open-source software tracking application used to track projects across different teams and revision levels.

Installing Git prompts you to select a text editor. If you don't have one, we strongly advise you to install it prior to installing Git.

- **Step 1: Download to the official website-** <https://git-scm.com/downloads>

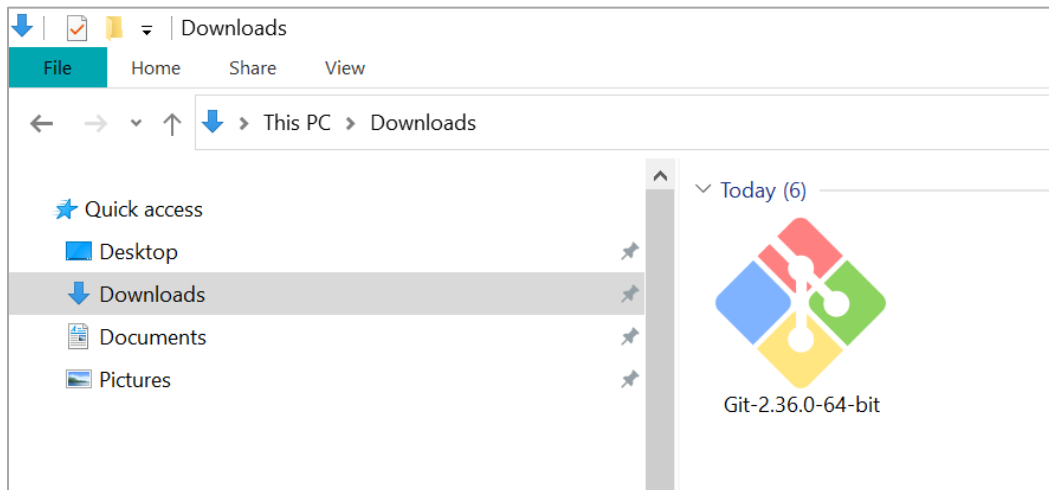
Click the download link for Windows and allow the download to complete.



- **Step 2: Extract and Launch Git Installer**

Browse to the download location (or use the download shortcut in your browser).

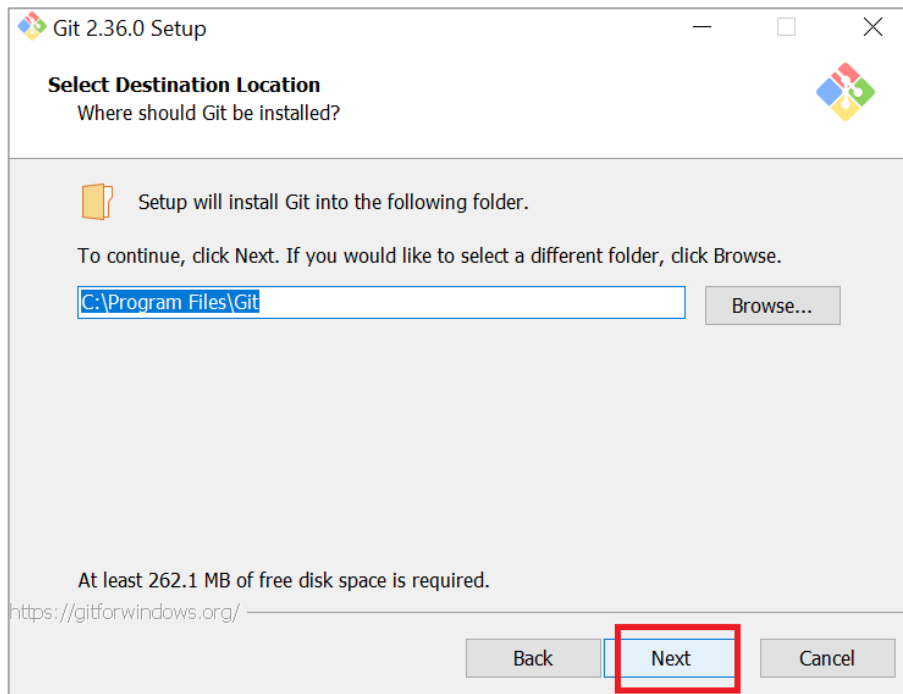
Double-click the file to extract and launch the installer.



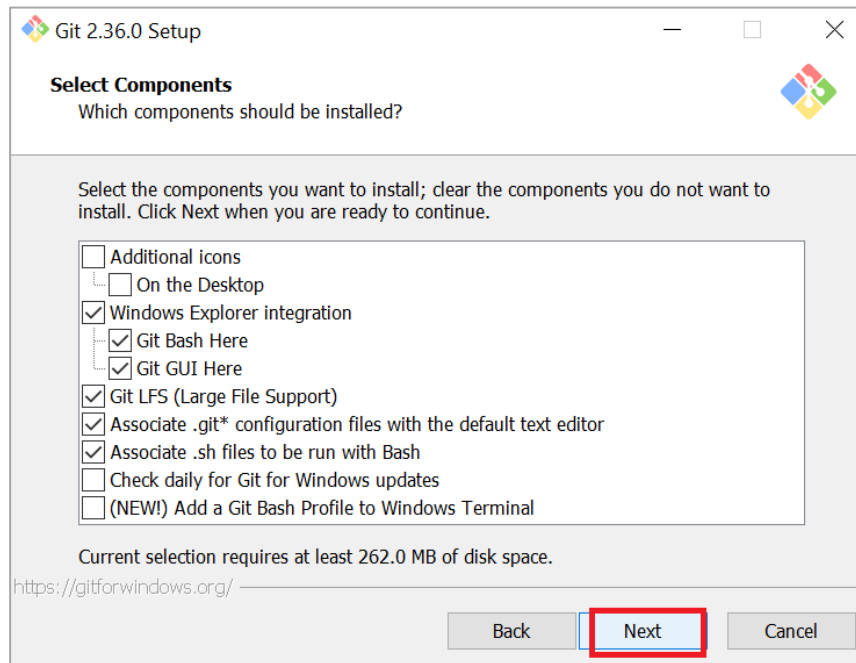
- **Step 3: Allow the app to make changes to your device by clicking Yes on the User Account Control dialog that opens.**
- **Step 4: Review the GNU General Public License, and when you're ready to install, click Next.**



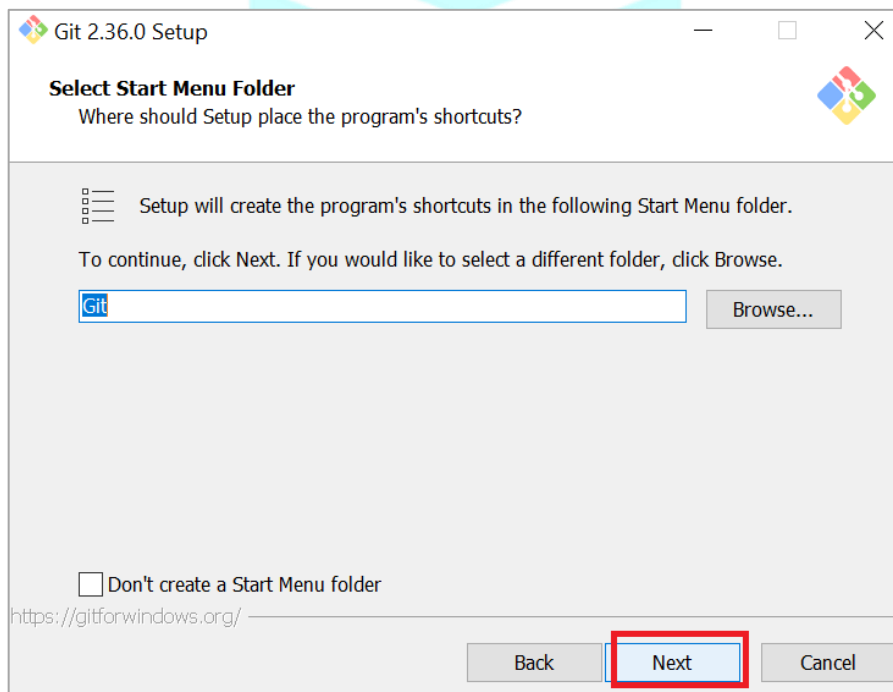
- **Step 5: The installer will ask you for an installation location. Leave the default, unless you have reason to change it, and click Next.**



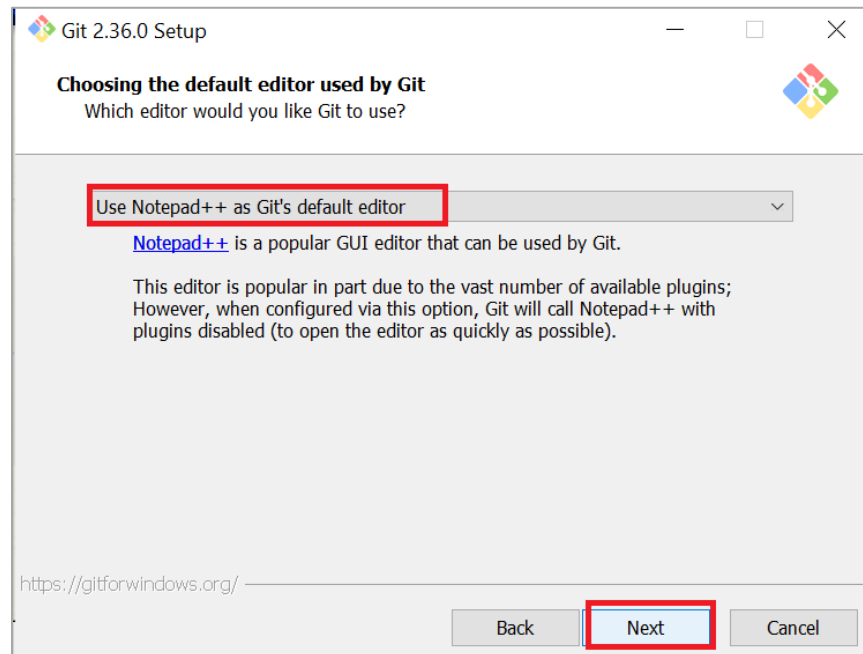
- **Step 6: A component selection screen will appear. Leave the defaults unless you have a specific need to change them and click Next.**



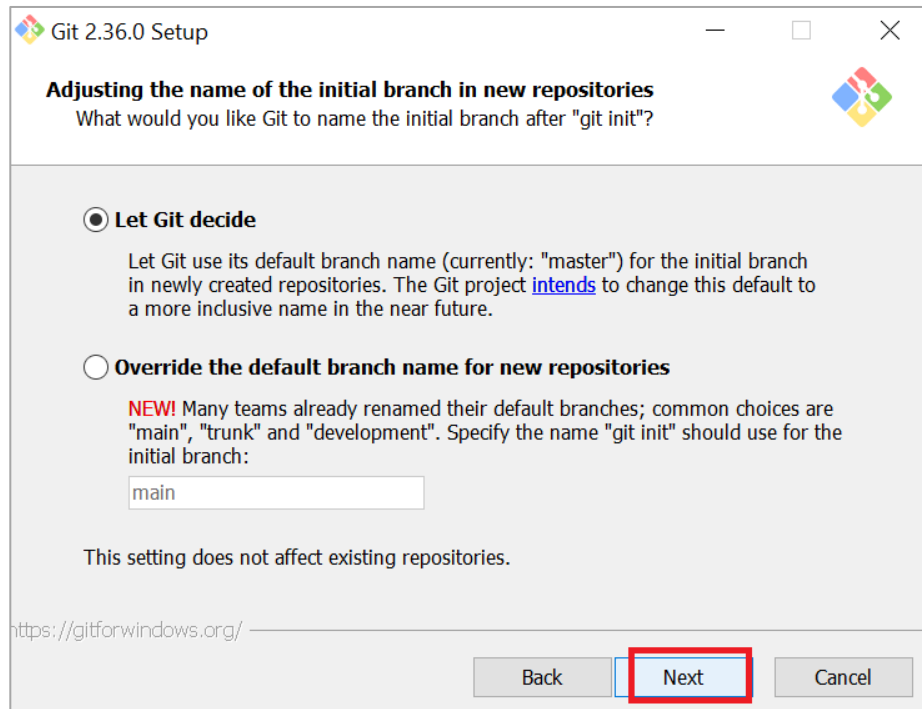
- **Step 7: The installer will offer to create a start menu folder. Simply click Next.**



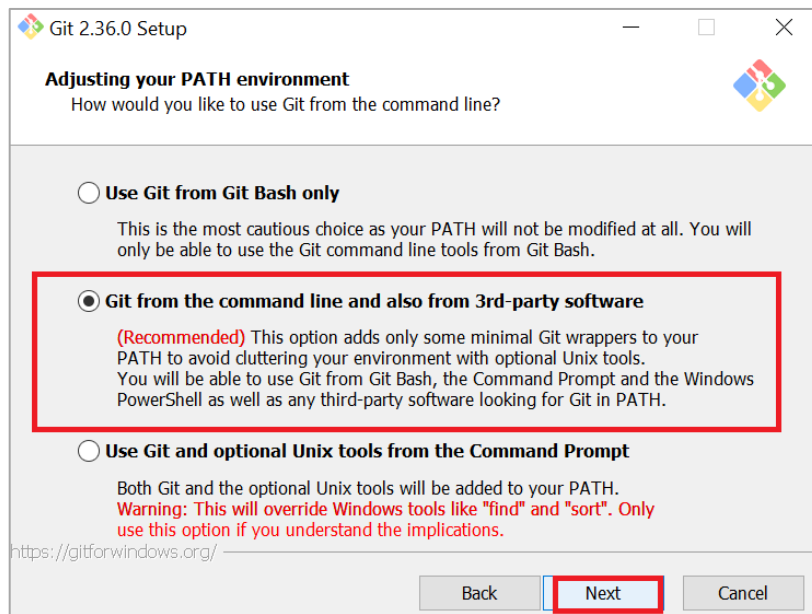
- **Step 8: Select a text editor you'd like to use with Git. Use the drop-down menu to select Notepad++ (or whichever text editor you prefer) and click Next.**



- **Step 9:** The next step allows you to choose a different name for your initial branch. The default is 'master.' Unless you're working in a team that requires a different name, leave the default option and click Next.

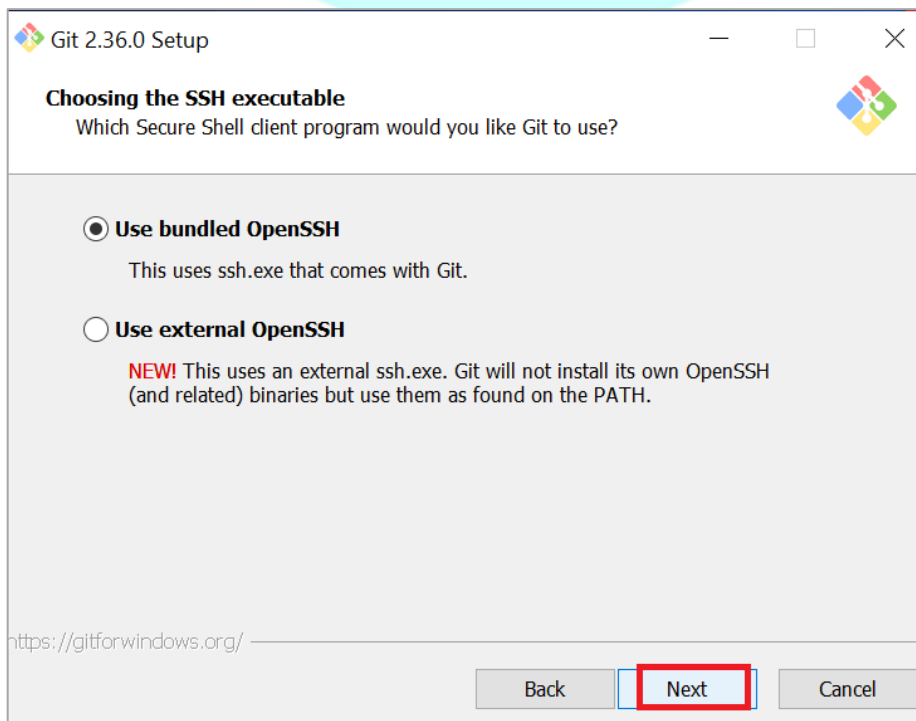


- **Step 10:** This installation step allows you to change the PATH environment. The PATH is the default set of directories included when you run a command from the command line. Leave this on the middle (recommended) selection and click Next.



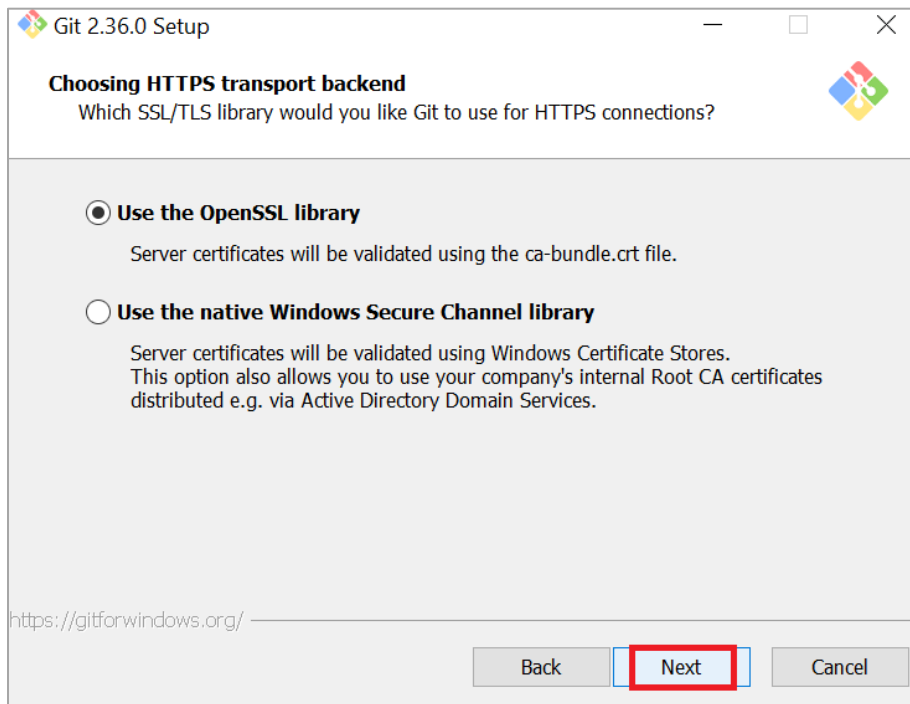
- **Step 11: Server Certificates, Line Endings and Terminal Emulators**

The installer now asks which **SSH client** you want Git to use. Git already comes with its own SSH client, so if you don't need a specific one, leave the default option and click **Next**.



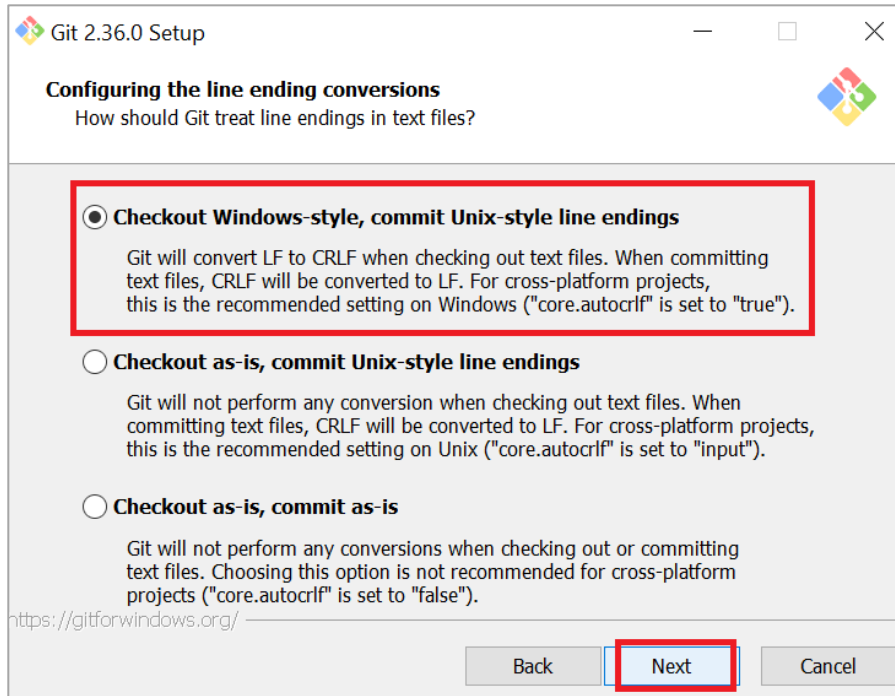
- **Step 12: This option is related to server certificates. Most users should use the default.**

If you're working in an Active Directory Environment, you may need to switch to Windows Store Certificates. Click Next.

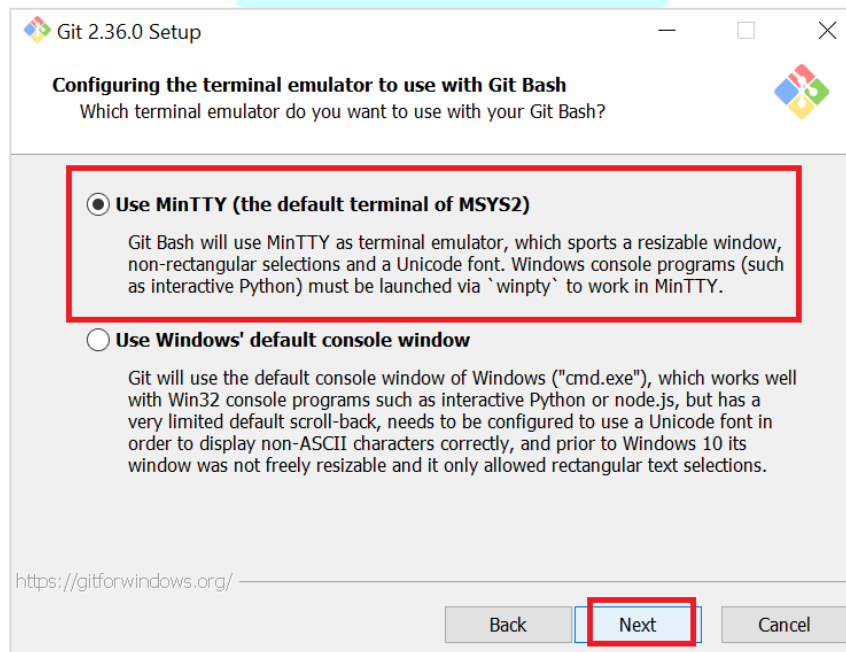


- **Step 13: The next selection converts line endings. It is recommended that you leave the default selection.**

If you're working in an Active Directory Environment, you may need to switch to Windows Store Certificates. Click Next.

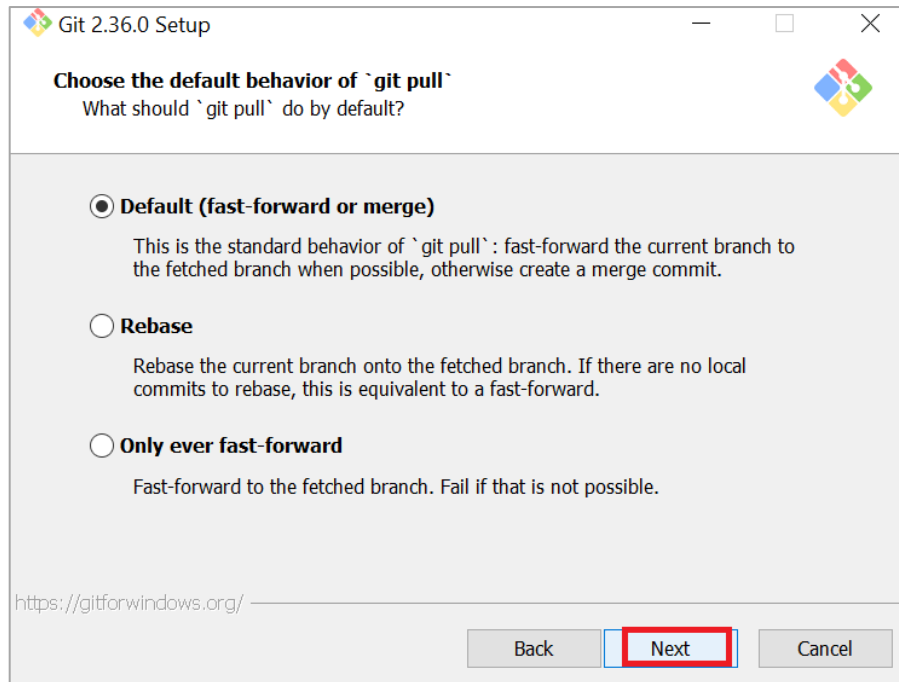


- **Step 14: Choose the terminal emulator you want to use. The default MinTTY is recommended, for its features. Click Next.**



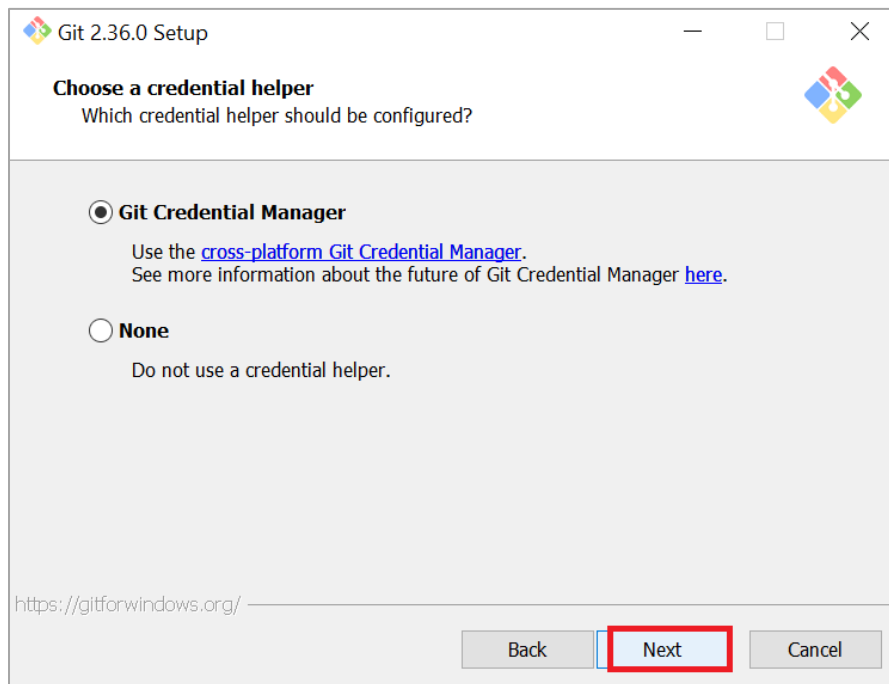
- **Step 15:** The installer now asks what the git pull command should do. The default option is recommended unless you specifically need to change its behavior.

Click Next to continue with the installation.



- **Step 16:** Next you should choose which credential helper to use. Git uses credential helpers to fetch or save credentials.

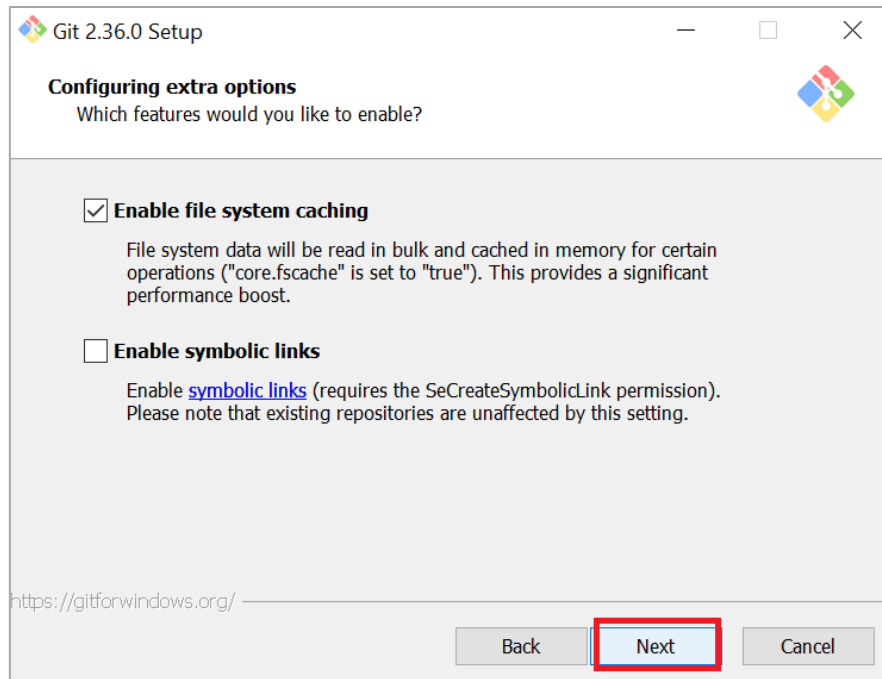
Leave the default option as it is the most stable one, and click Next.



- **Step 17: Additional Customization Options**

The default options are recommended;

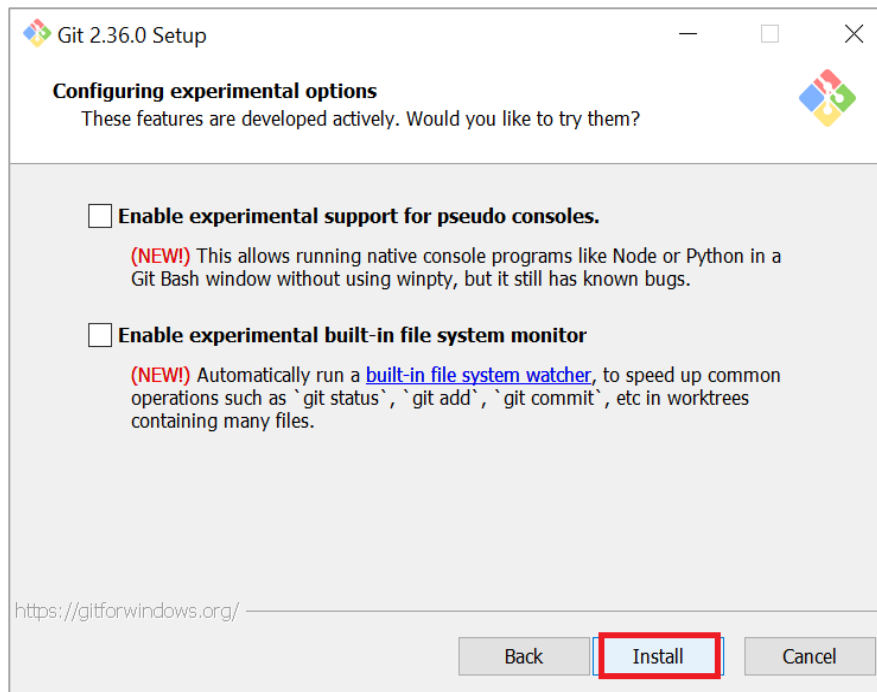
However this step allows you to decide which extra option you would like to enable. If you use symbolic links, which are like shortcuts for the command line, tick the box. Click **Next**.



- **Step 18: Depending on the version of Git you're installing; it may offer to install experimental features.**

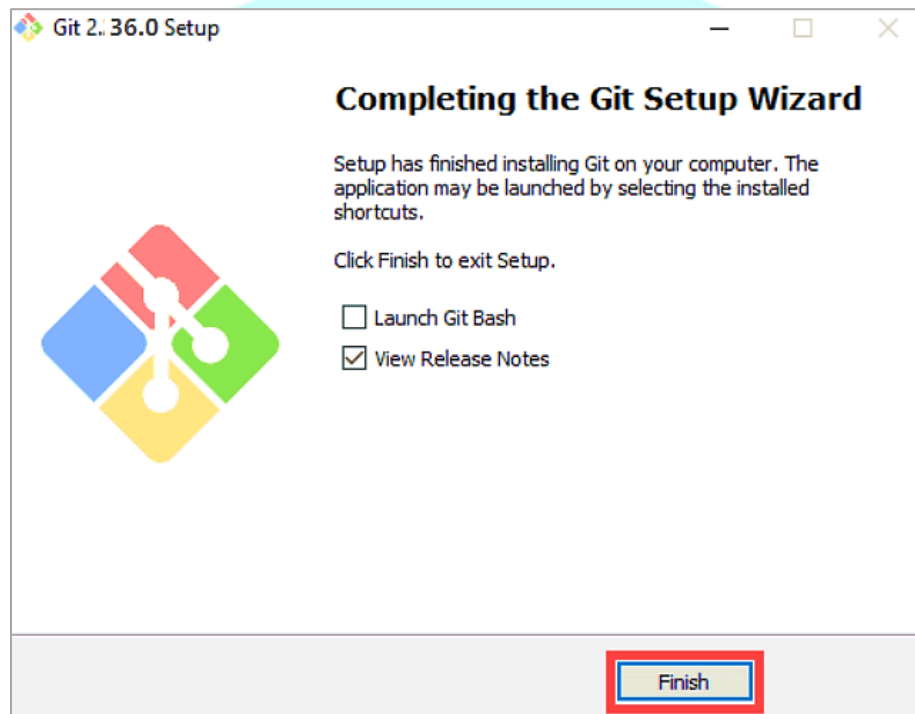
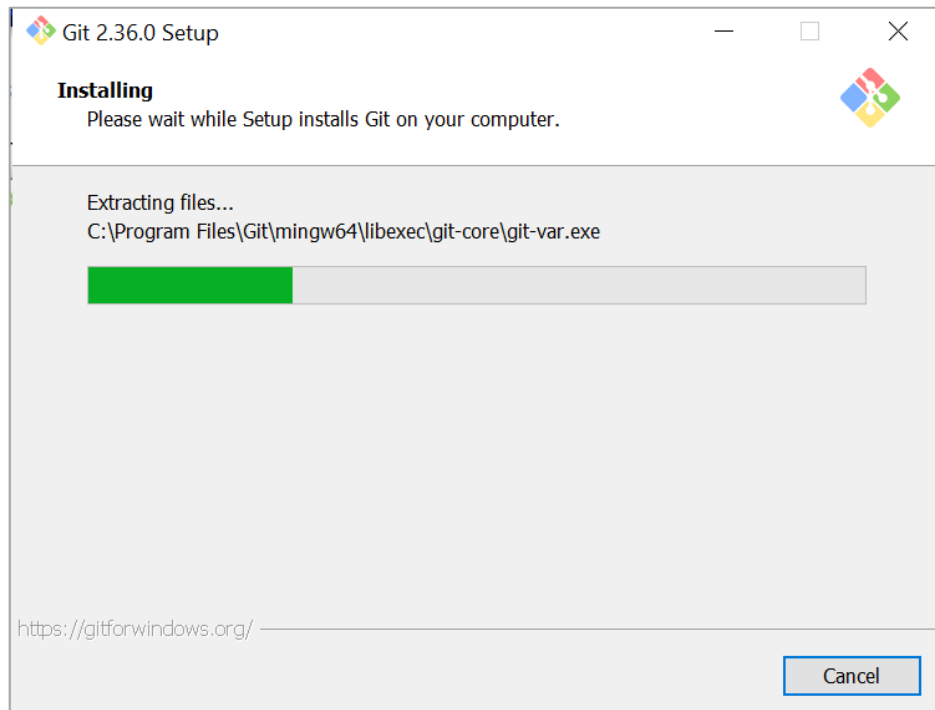
At the time this article was written, the options to include support for pseudo controls and a built-in file system monitor were offered.

Unless you are feeling adventurous, leave them unchecked and click Install.



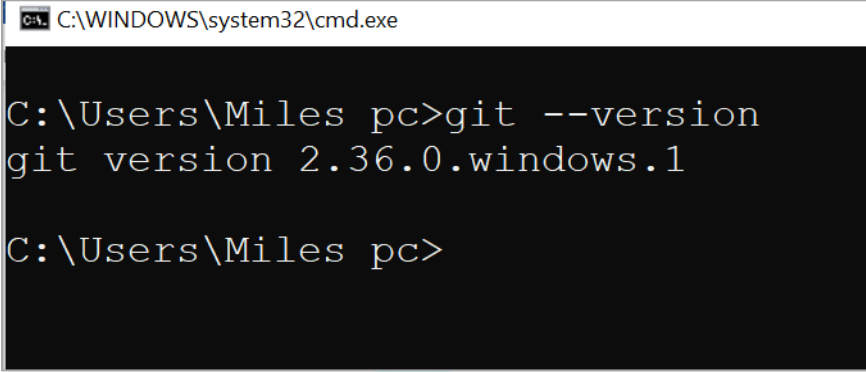
- **Step 19: Complete Git Installation Process**

Once the installation is complete, tick the boxes to view the Release Notes or Launch Git Bash, then click **Finish**.



- **Step 20: Verify the Git version**

To check if Git has successfully installed on your system, run **git --version** command on the command prompt.



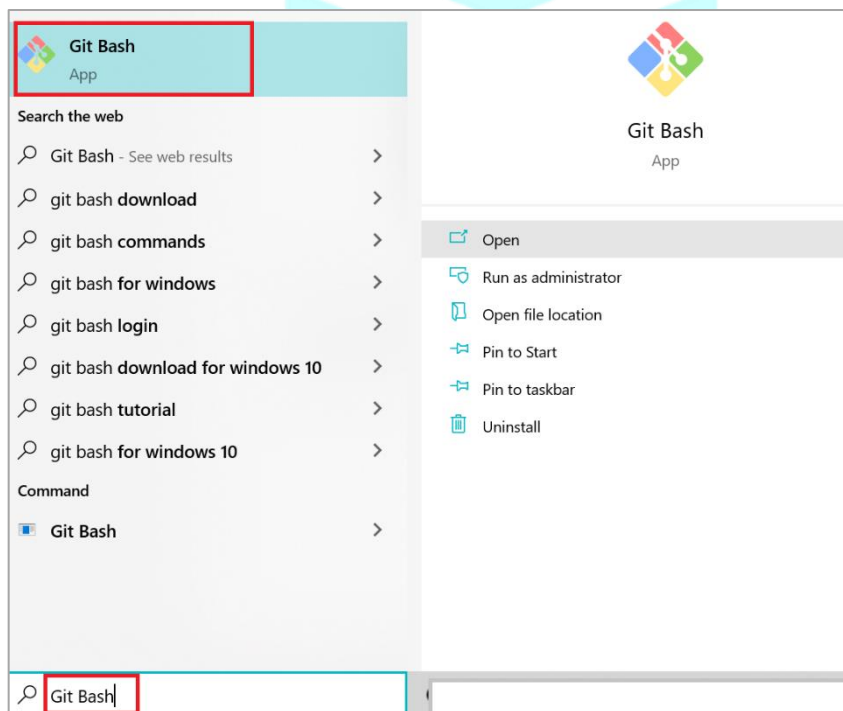
```
C:\WINDOWS\system32\cmd.exe

C:\Users\Miles pc>git --version
git version 2.36.0.windows.1

C:\Users\Miles pc>
```

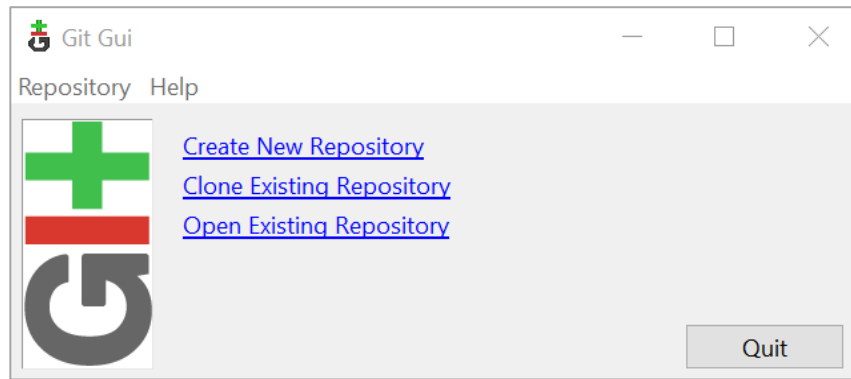
- **Step 21: Launch Git Bash Shell**

To launch **Git Bash** open the **Windows Start** menu, type *git bash* , and press **Enter** (or click the application icon).



- **Step 22: Launch Git GUI**

To launch **Git GUI** open the **Windows Start** menu, type **git gui** and press **Enter** (or click the application icon).



You're all set!

8. Microsoft Azure Account Setup

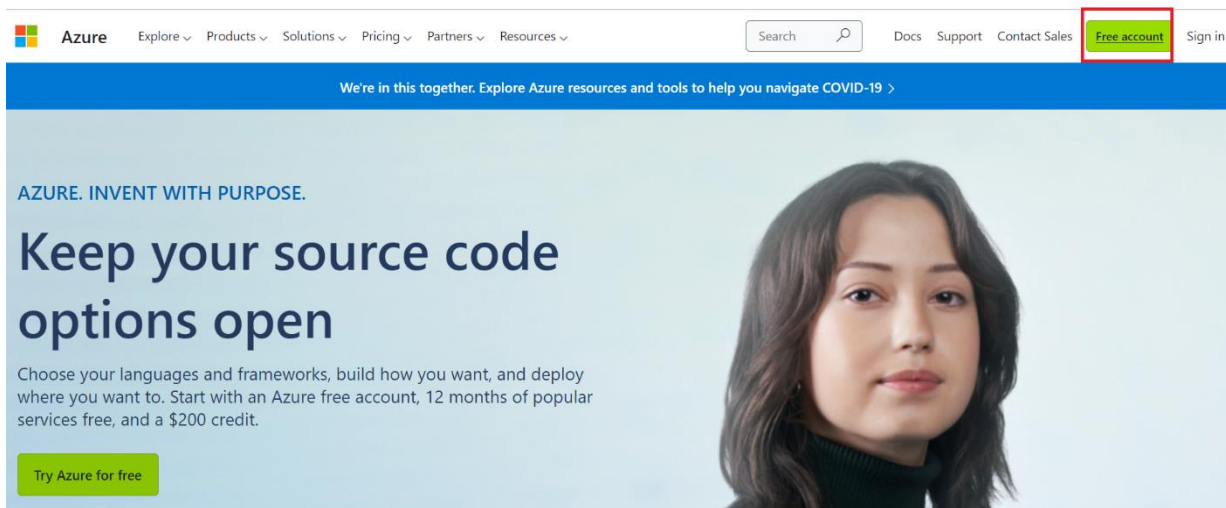
You can create a free Azure account. Azure allows you to create a free trial account for 30 days and gives you some credit of 200 USD to use azure services. After completing your trial of 30 days and using your credits, you can upgrade your subscription and keep using the paid services.

You can use your personal email address or Microsoft outlook account to create an account on the Azure platform. But you cannot sign up here with your work or school account.

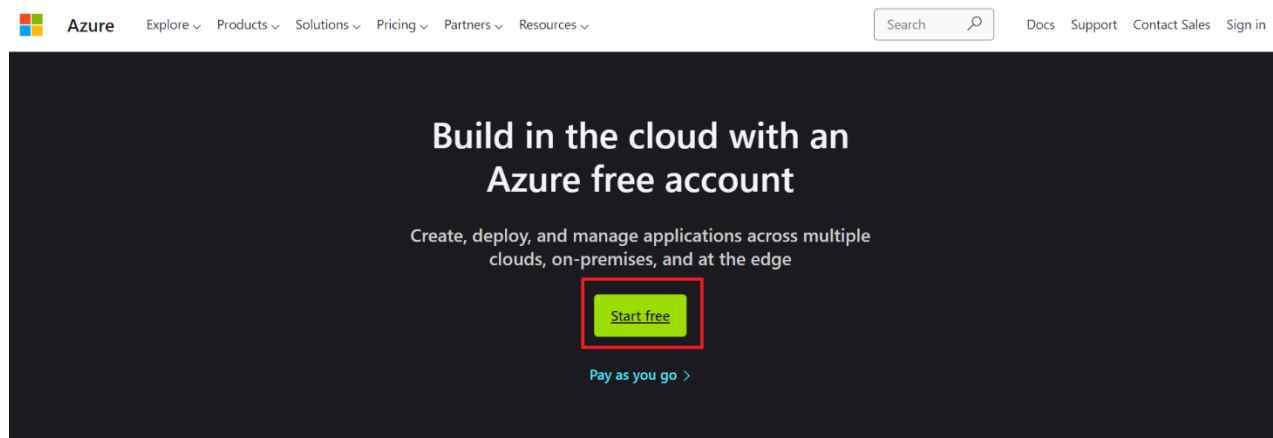
Now, we will see how to create a free account on Azure.

- **Step 1: Visit the Azure Home Page - <https://azure.microsoft.com/en-us/>**

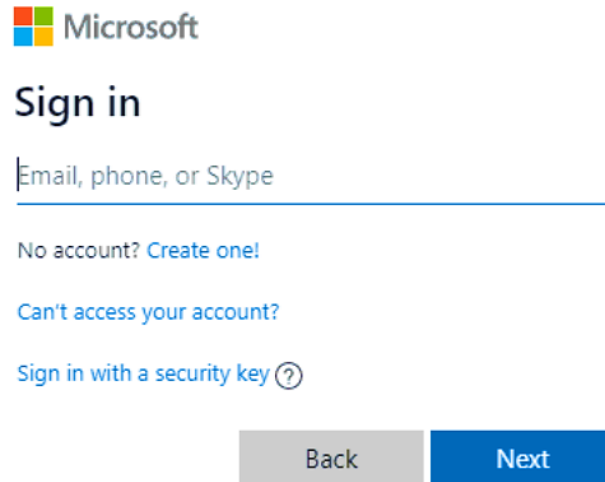
- **Step 2: Click on the FREE ACCOUNT as shown in the below image.**



- **Step 3: Click on START FREE.**



- **Step 4: Sign in using an email address or create a new one.**



The image shows the Microsoft sign-in page. At the top is the Microsoft logo. Below it is the heading "Sign in". Under the heading is a text input field with the placeholder text "Email, phone, or Skype". Below the input field are three links: "No account? Create one!", "Can't access your account?", and "Sign in with a security key ?". At the bottom of the form are two buttons: a grey "Back" button and a blue "Next" button.

- **Step 5: Now enter the following details.**

- Country/Region
- Your Name
- Email Address
- Mobile Number
- And verify the mobile number

Your profile

Country/Region ⓘ
India ▼

Choose the location that matches your billing address. You cannot change this selection later. If your country is not listed, the offer is not available in your region. [Learn More](#)

First name
[]

Middle name (Optional)
[]

Last name
[]

Email address ⓘ
[]

Phone ⓘ
Example: 81234-56789

☐ Use a different phone number to verify your identity.

Text me **Call me**

Create your Azure free account

- Popular services free for 12 months
- 25+ services always free
- ₹13,300 credit to use in your first 30 days

No automatic charges
After your credit is over, we'll ask you if you want to continue with pay-as-you-go. If you do, you'll only pay if you use more than the free amounts of services.

- **Step 6: Then it will ask you to provide the Phone number details, make sure to provide a valid phone number. Because you need to confirm by a text message or by a call.**

Once verification is done, click on Next.

Identity verification by phone

A text or phone call helps us make sure this is you.

Country code

India (+91) ▼

Phone number

[]

Text me

Call me

Verification code

[]

Verify code

- **Step 7: Now you need to provide the credit card details. Provide everything correctly here. Because it is going to deduct a little amount of money from your account.**

signup.azure.com/signup?offer=ms-azr-0044p&appId=1028ref=azureplat-generic&redirectURL=https%3F%2Fazure.microsoft.com%2Fen-us%2Fget-started%2Fwelcome-to-azure%2F

You won't be charged unless you upgrade.

We accept the following cards:

VISA  

Cardholder Name
Atul Kumar

Card number

Expires
MM YY

CVV
 What is a CVV?

Address line 1

Address line 2 (Optional)

City

County (Optional)

Postal Code

Country/Region

30 days—beyond the free product amounts.

- ✓ **25+ always-free products**
Take advantage of more than 25 products, including *serverless*, *containers*, and *artificial intelligence*, that are always free. Get these in your first 30 days, and always—once you choose to upgrade.
- ✓ **No automatic charges**
You won't be charged unless you choose to upgrade. Before the end of your first 30 days, you'll be notified and have the chance to upgrade and start paying only for the resources you use beyond the free amounts.

- **Step 8: Check the Terms and Conditions and click Sign-Up.**

Follow these steps to get started. We ask for these details to protect your account and information. There are no upfront charges or fees.

- About you
- Identity verification by phone
- Identity verification by card
- Agreement**

☒ I agree to the [subscription agreement](#), [offer details](#), and [privacy statement](#).

☒ I would like information, tips, and offers from Microsoft or selected partners about Azure, including Azure Newsletter, Pricing updates, and other Microsoft products and services.

Sign up

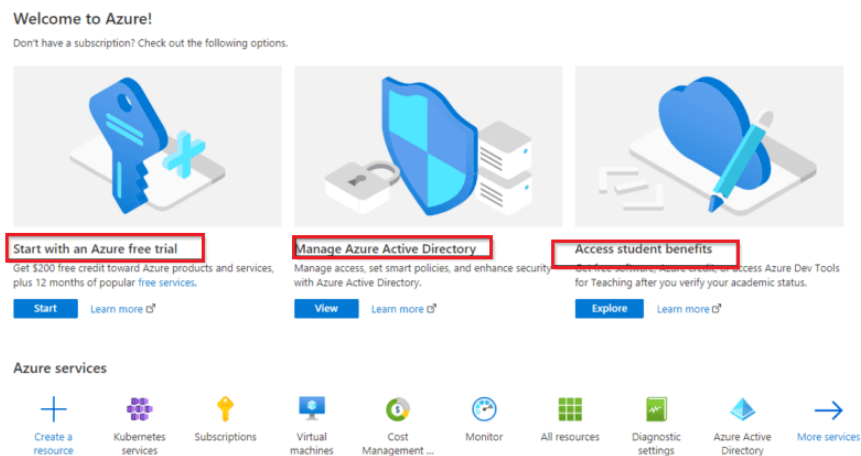
What's included

- ✓ **12 months of free products**
Get free access to popular products like *virtual machines*, *storage*, and *databases* in your first 30 days, and for 12 months after you upgrade your account to pay-as-you-go pricing.
- ✓ **£150 credit**
Use your £150 credit to experiment with any Azure service in your first 30 days—beyond the free product amounts.
- ✓ **25+ always-free products**
Take advantage of more than 25 products, including *serverless*, *containers*, and *artificial intelligence*, that are always free. Get these in your first 30 days, and always—once you choose to upgrade.

- **Step 9: You have successfully created a Microsoft Azure free account.**

- **Step 10:** Click on Portal on the top right corner of the screen. You'll be redirected to the Azure Portal.
- **Step 11:** If you have exhausted your free credit then you have to move to the Pay as you go subscription policy.

if you have crossed the limit or time limit then you will get “your subscription is disabled and cannot perform operations until its re-enabled”



You cannot create multiple Microsoft azure free accounts or after one account expires to another account for free credit of 200\$ using a single Credit card.

Note: *If you are a student and don't have a credit card, then you can also register using your student ID.*

```
Select Command Prompt
C:\Users\Miles pc>docker version
Client:
Cloud integration: v1.0.22
Version: 20.10.13
API version: 1.41
Go version: go1.16.15
Git commit: a224086
Built: Thu Mar 10 14:13:04 2022
OS/Arch: windows/amd64
Context: default
Experimental: true

Server: Docker Desktop 4.6.0 (75818)
Engine:
Version: 20.10.13
API version: 1.41 (minimum version 1.12)
Go version: go1.16.15
Git commit: 906f57f
Built: Thu Mar 10 14:06:05 2022
OS/Arch: linux/amd64
Experimental: false
containerd:
Version: 1.5.10
GitCommit: 2a1d4dbdb2a1030dc5b01e96fb110a9d9f150ecc
runc:
Version: 1.0.3
GitCommit: v1.0.3-0-gf46b6ba
docker-init:
Version: 0.19.0
```

Contact Us

If you are still facing any issues while installing the software and configuring the system, please free to reach out to us at **+91 9999 123 502/503** or email us at hello@scholarhat.com